

Matlacha Pass Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

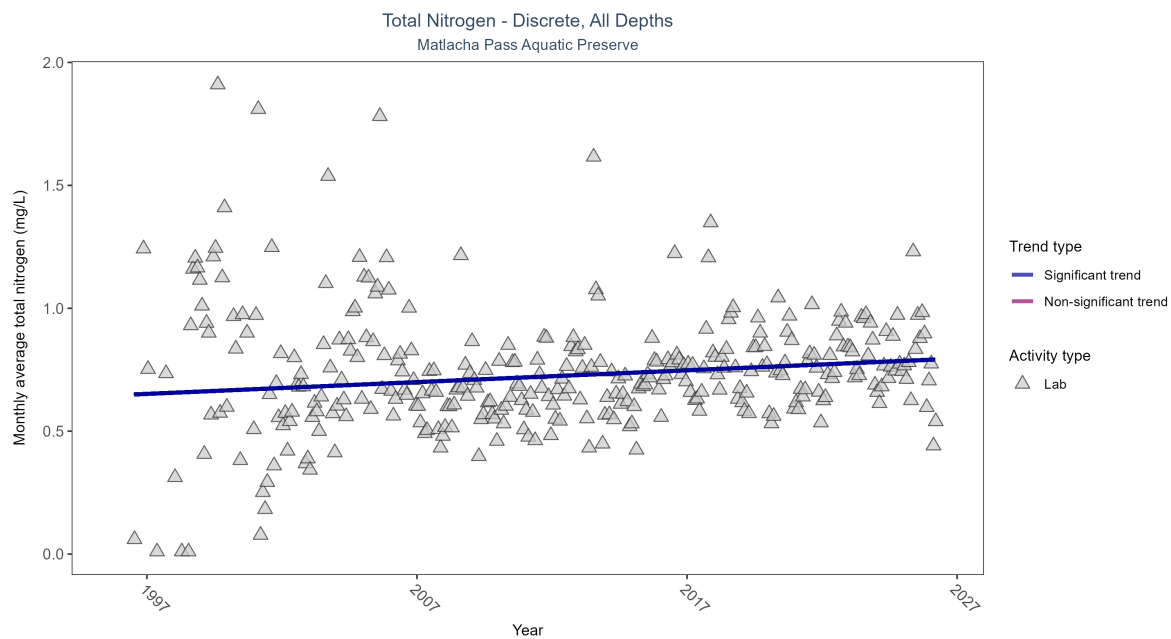


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2168	31	1996 - 2026	0.71	0.14705	0.64684	0.00479	0.0001

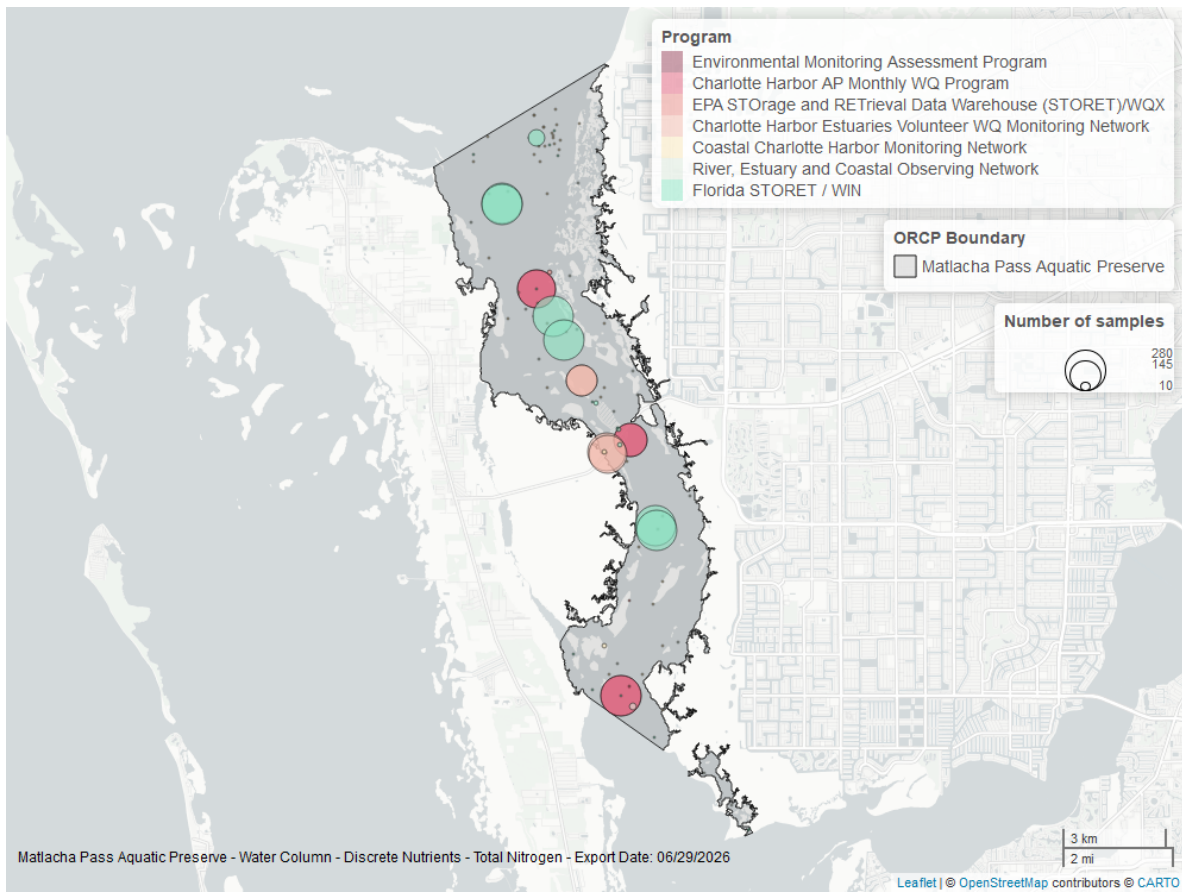


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

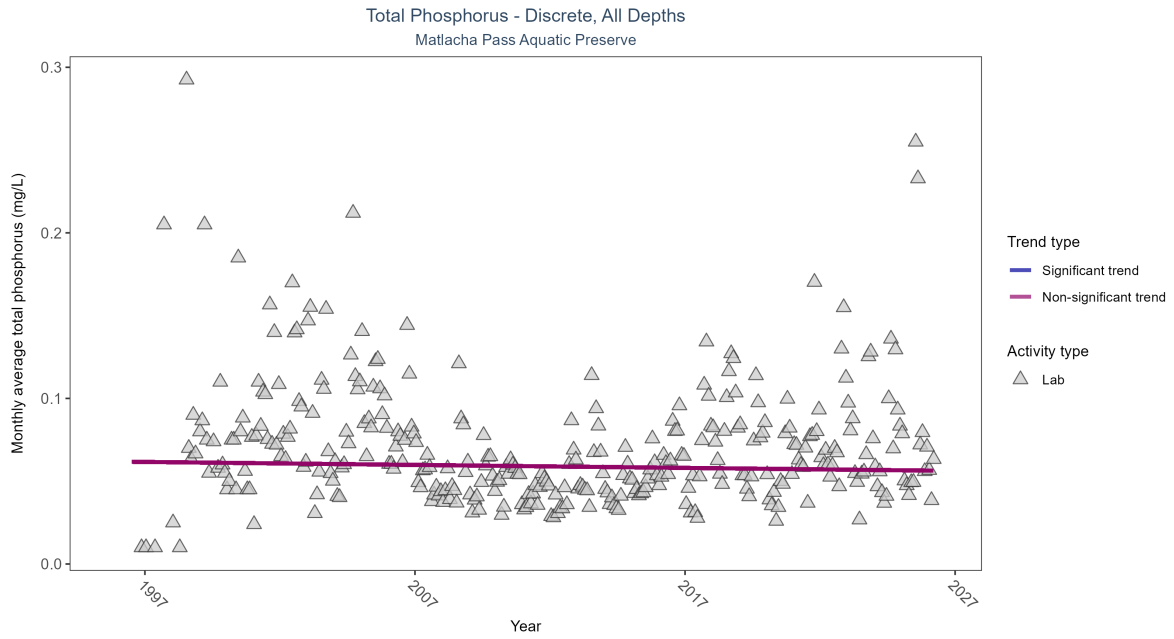


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1996 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2647	31	1996 - 2026	0.059	-0.03803	0.06182	-0.00018	0.3766

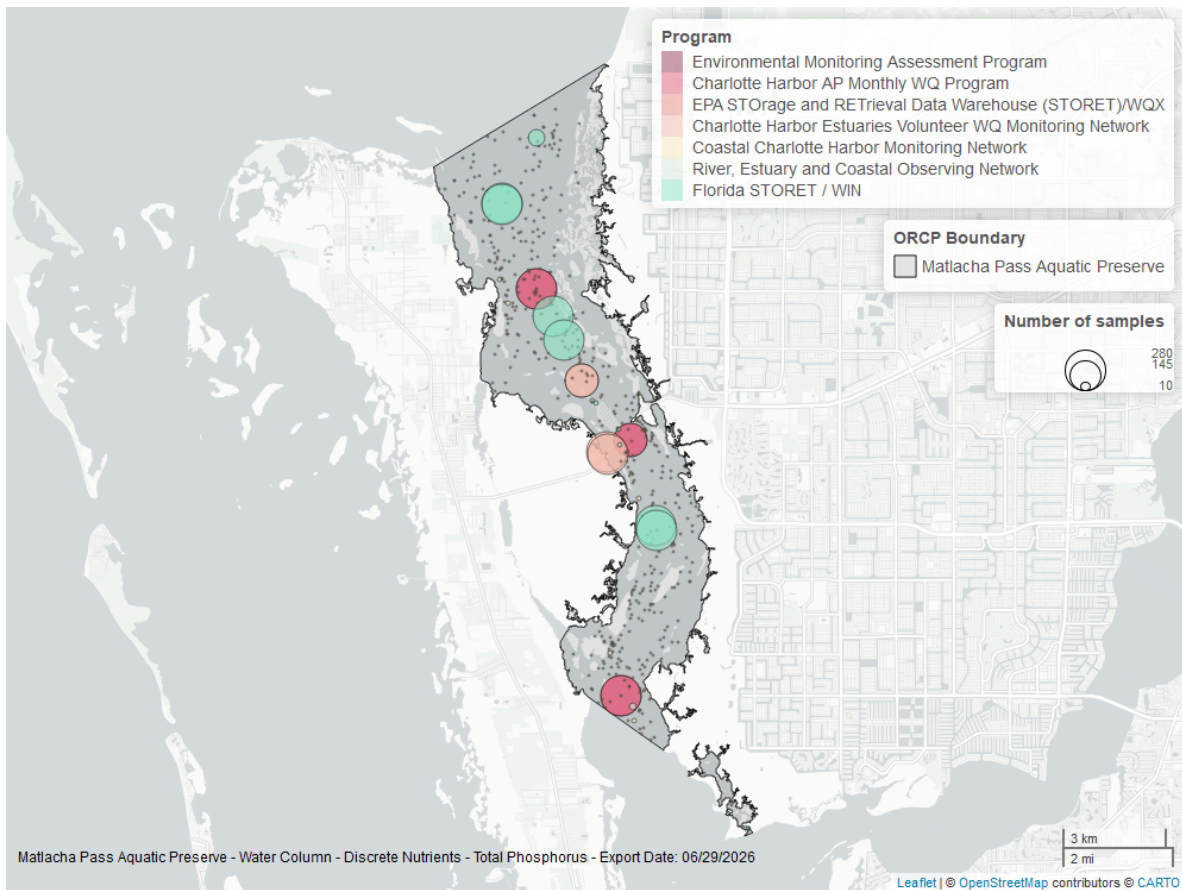


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

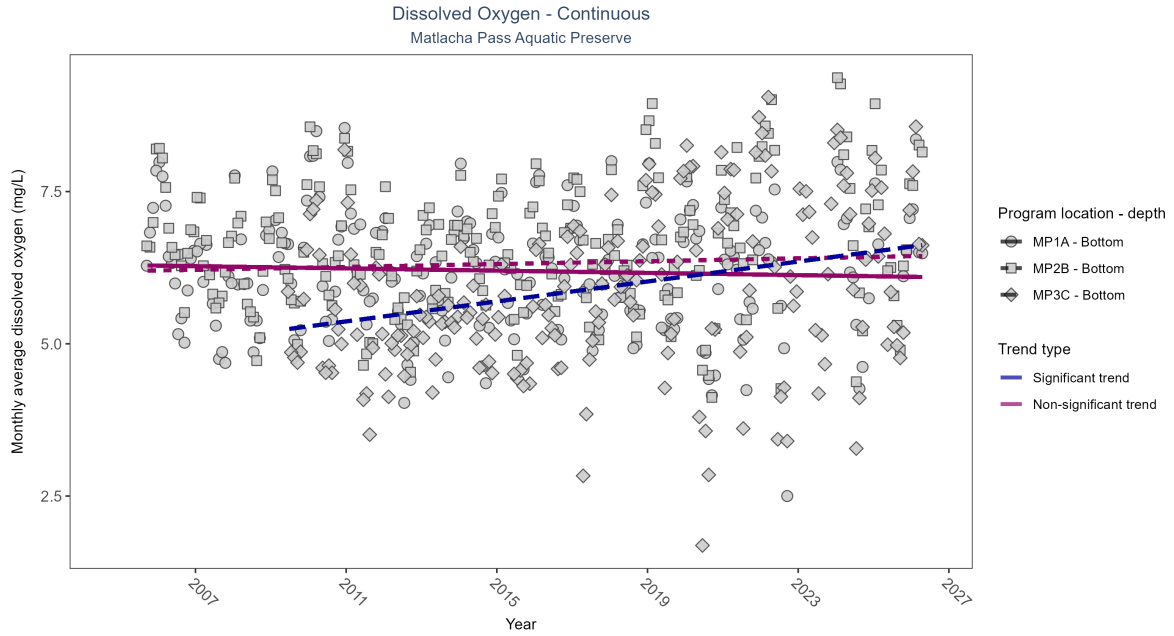


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At one program location, monthly average dissolved oxygen increased by 0.08 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	No detectable trend	571895	21	2005 - 2026	6.3	-0.06	6.29	-0.01	0.23
MP2B	No detectable trend	596589	21	2005 - 2026	6.7	0.08	6.19	0.01	0.08
MP3C	Increasing trend	511399	18	2009 - 2026	5.9	0.27	5.20	0.08	0.00

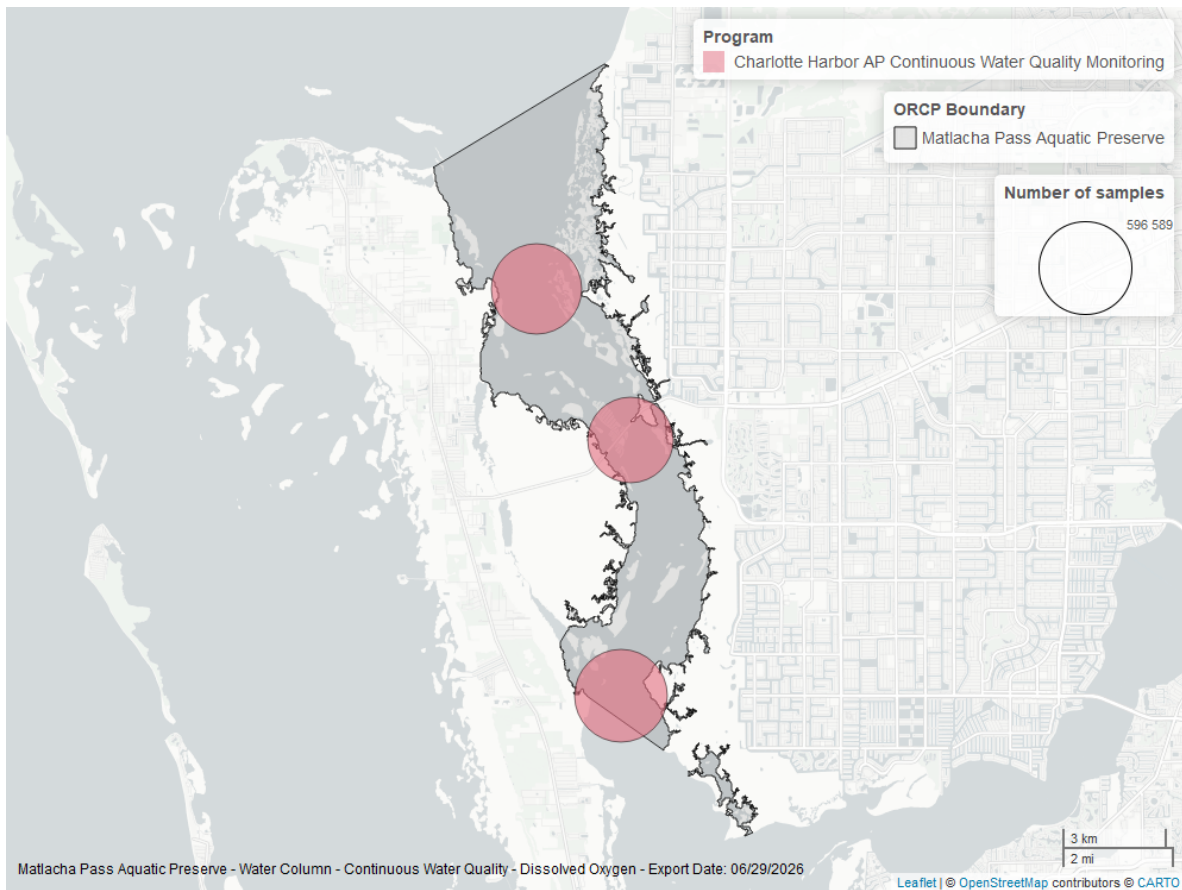


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

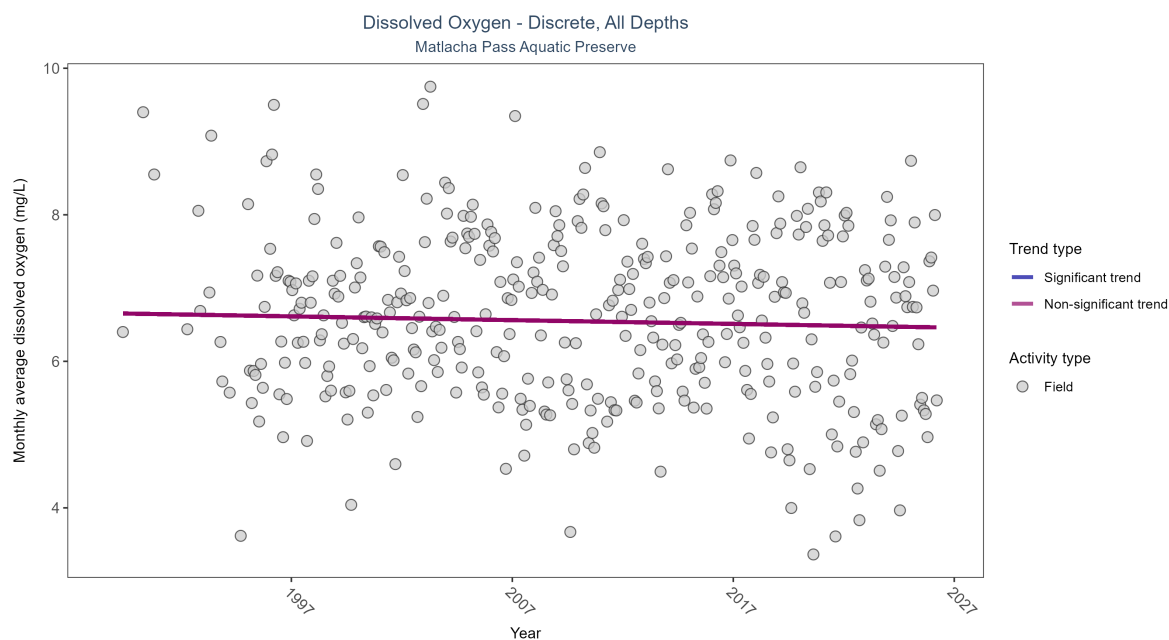


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1989 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	11940	37	1989 - 2026	6.6	-0.04036	6.65461	-0.00514	0.2699

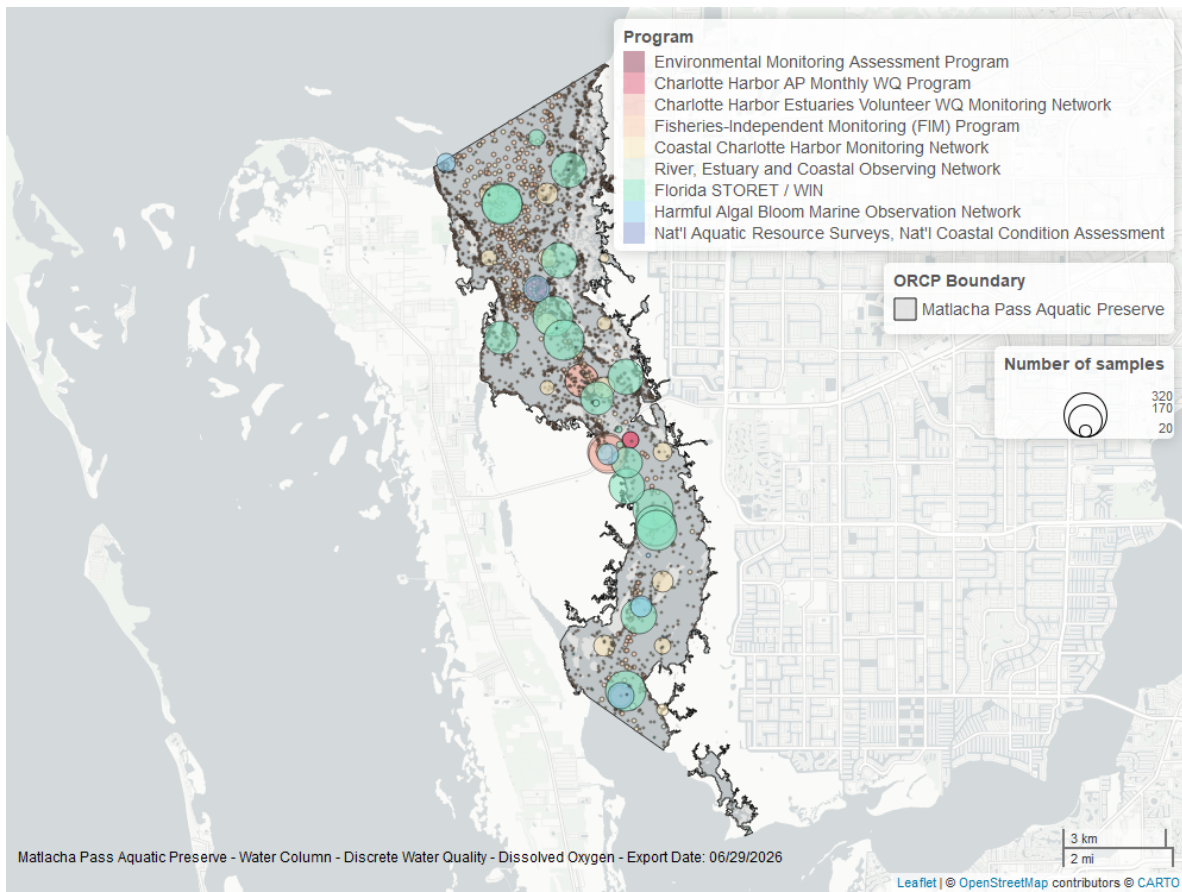


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

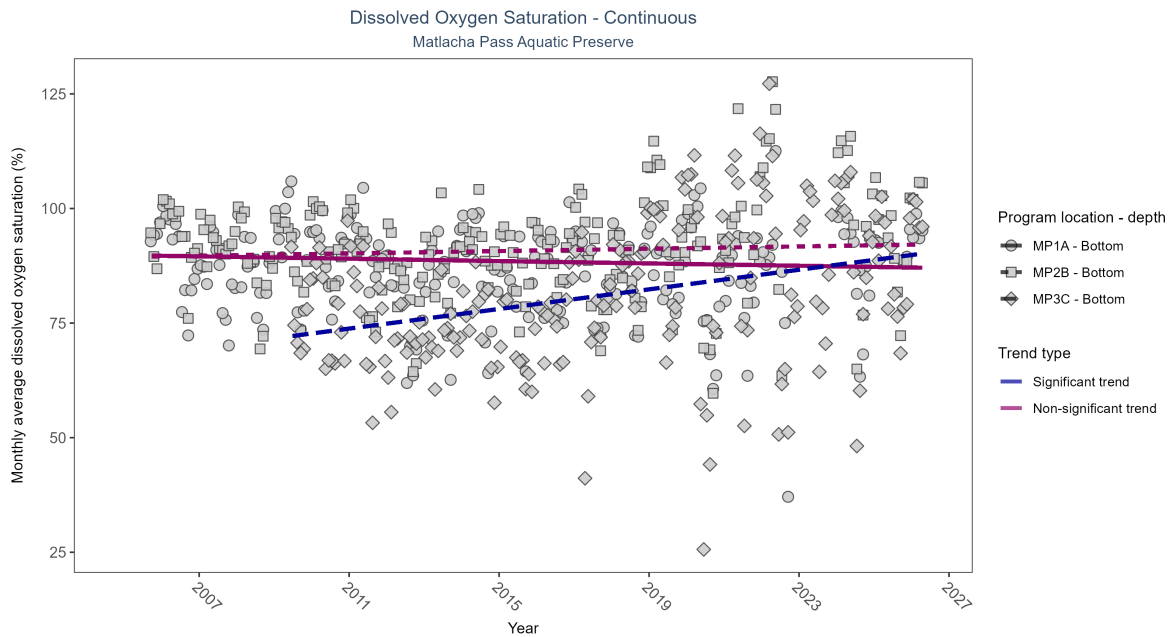


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At one program location, monthly average dissolved oxygen saturation increased by 1.07% per year. No detectable change in monthly average dissolved oxygen saturation was observed at two locations.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	No detectable trend	569253	21	2005 - 2026	88.3	-0.10	89.81	-0.13	0.06
MP2B	No detectable trend	596890	21	2005 - 2026	91.6	0.06	89.44	0.13	0.19
MP3C	Increasing trend	515527	18	2009 - 2026	81.9	0.27	71.66	1.07	0.00

\end{table}

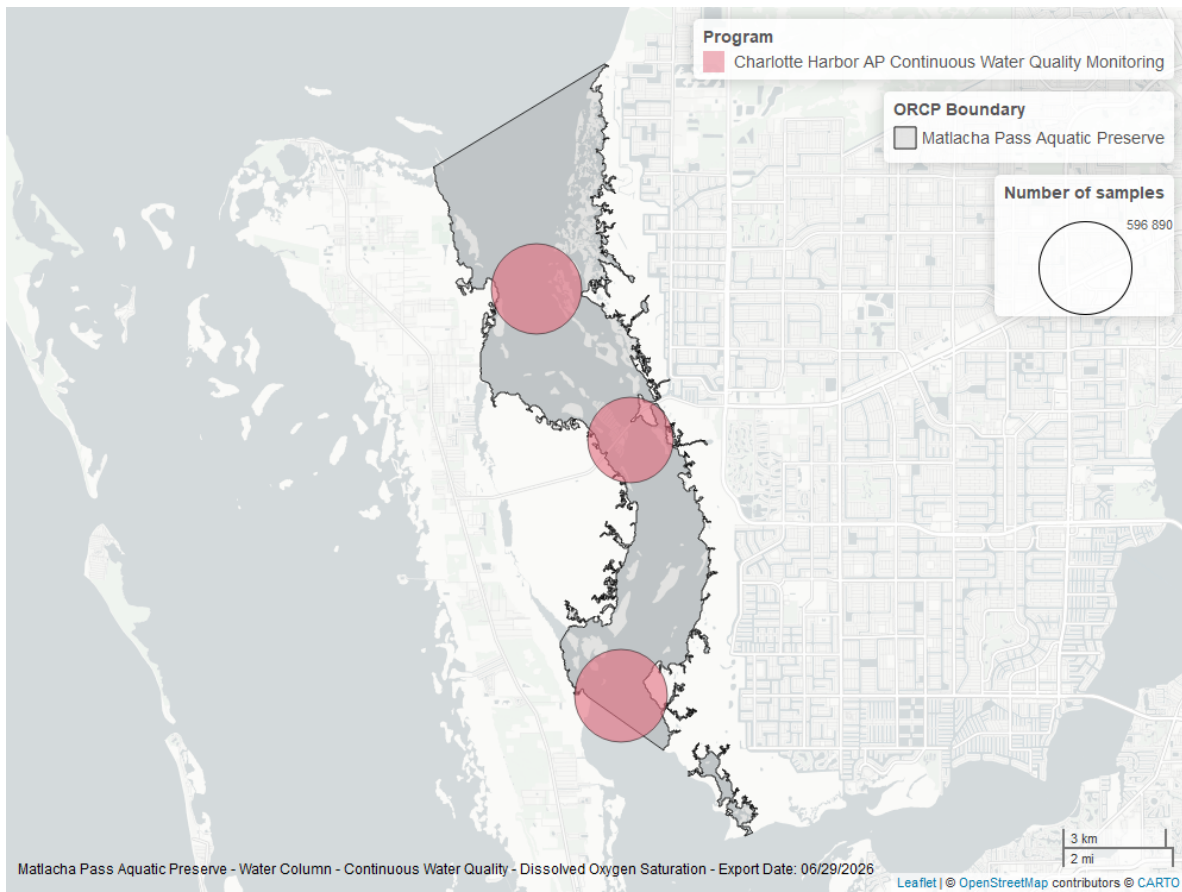


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

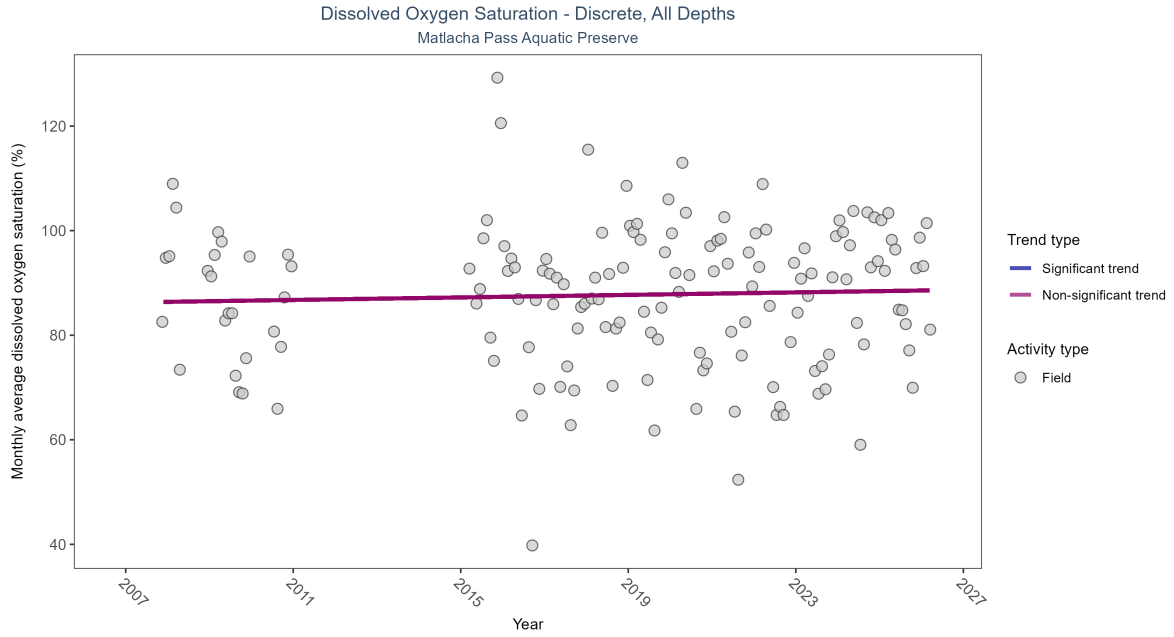


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 5: Dissolved oxygen saturation showed no detectable trend between 2007 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	1494	16	2007 - 2026	89.95	0.05712	86.25973	0.12043	0.3371

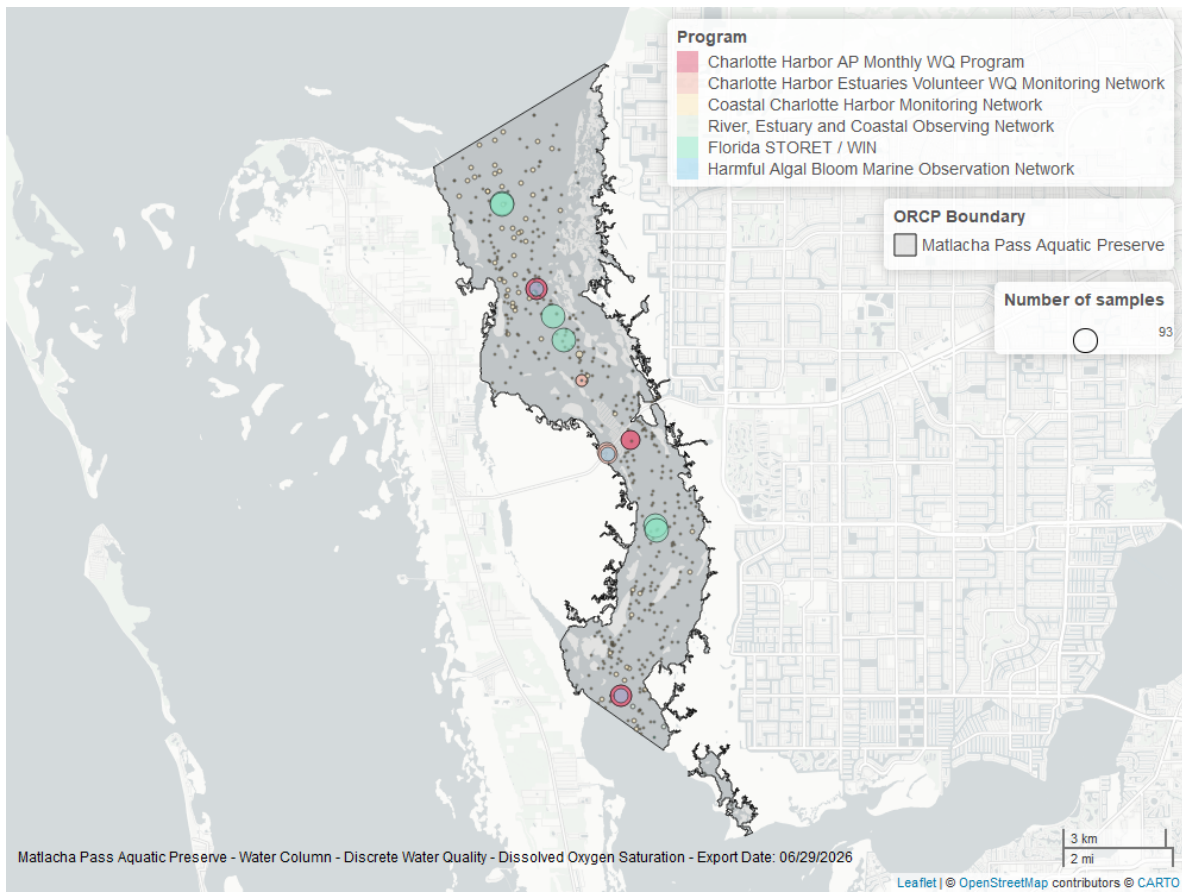


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

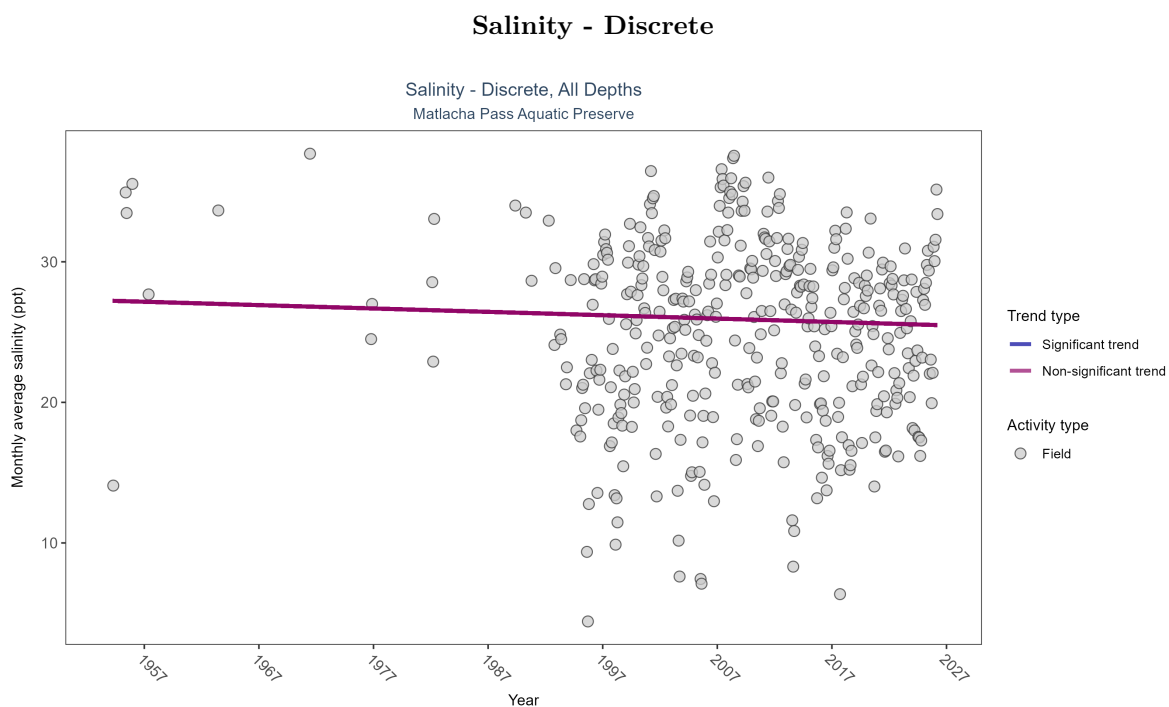


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Salinity showed no detectable trend between 1954 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	11268	44	1954 - 2026	25.9	-0.02997	27.23002	-0.02397	0.3728

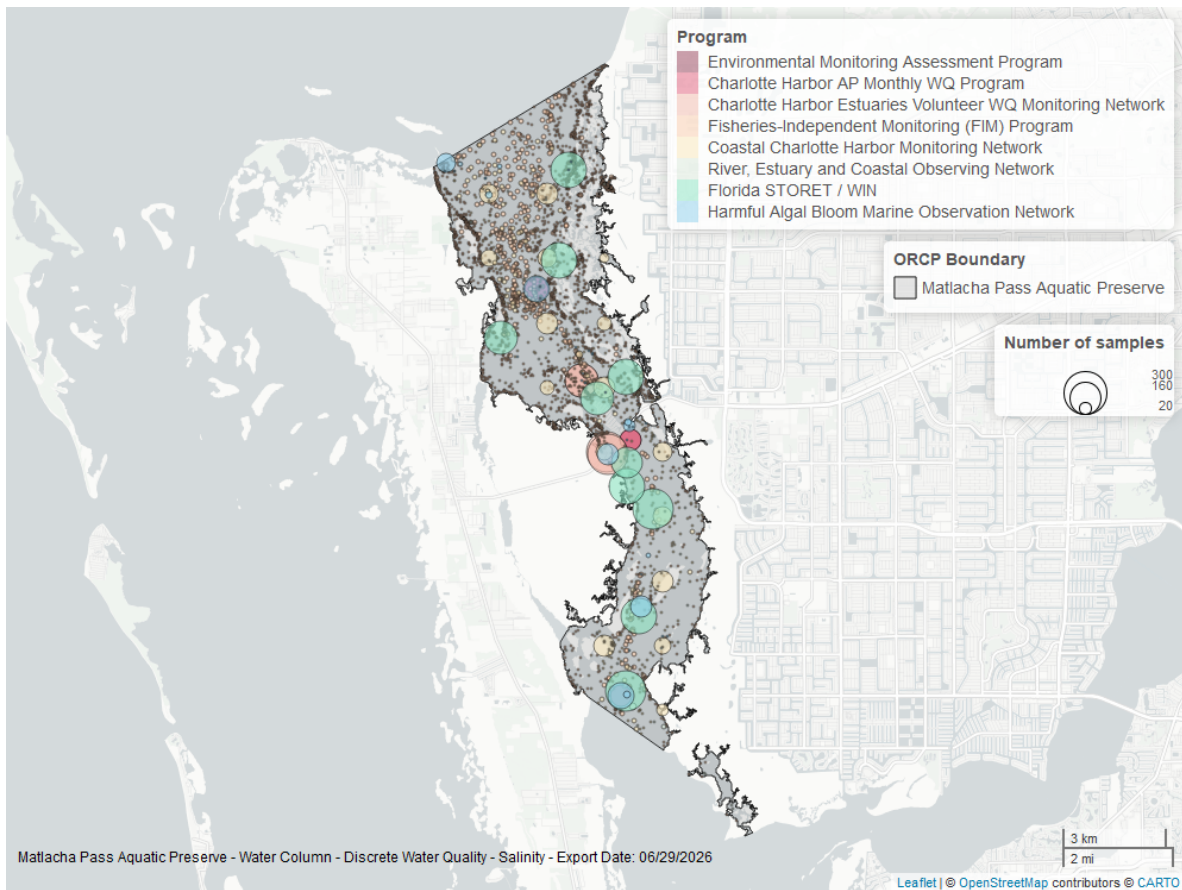


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

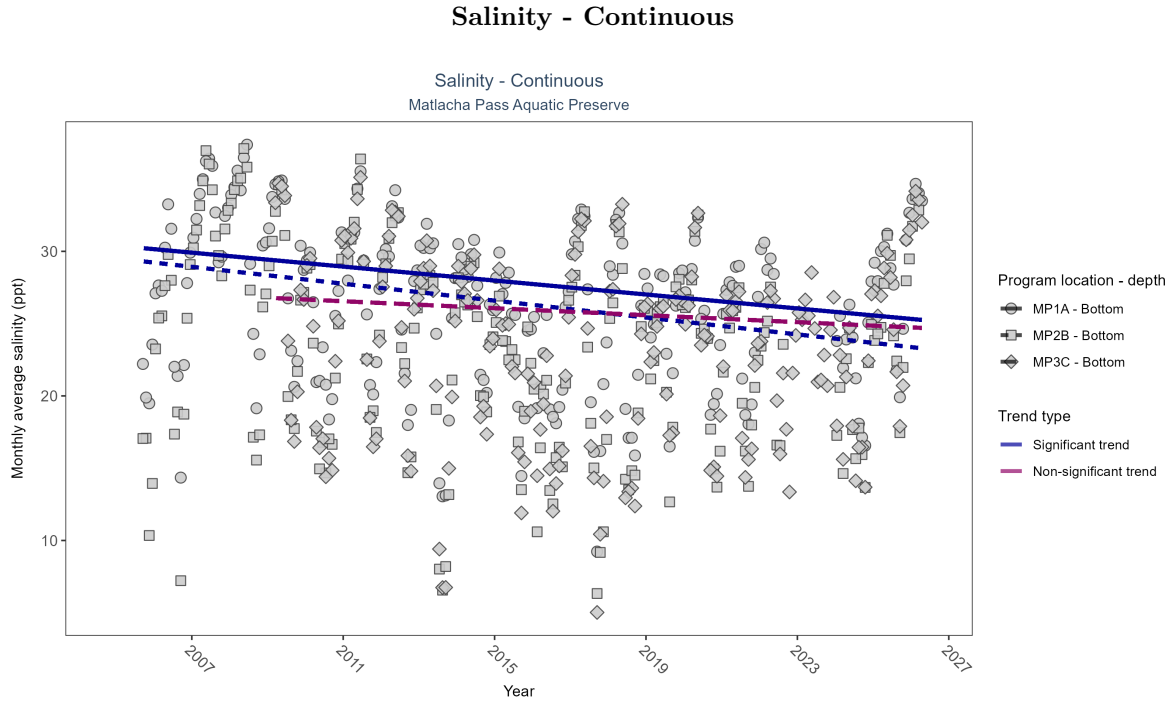


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At two program locations, monthly average salinity decreased by 0.24 ppt per year at one site and by 0.29 ppt per year at the other. No detectable change in monthly average salinity was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Decreasing trend	611503	21	2005 - 2026	27.2	-0.24	30.39	-0.24	0.00
MP2B	Decreasing trend	619663	21	2005 - 2026	24.5	-0.25	29.51	-0.29	0.00
MP3C	No detectable trend	549239	18	2009 - 2026	23.9	-0.08	26.79	-0.12	0.13

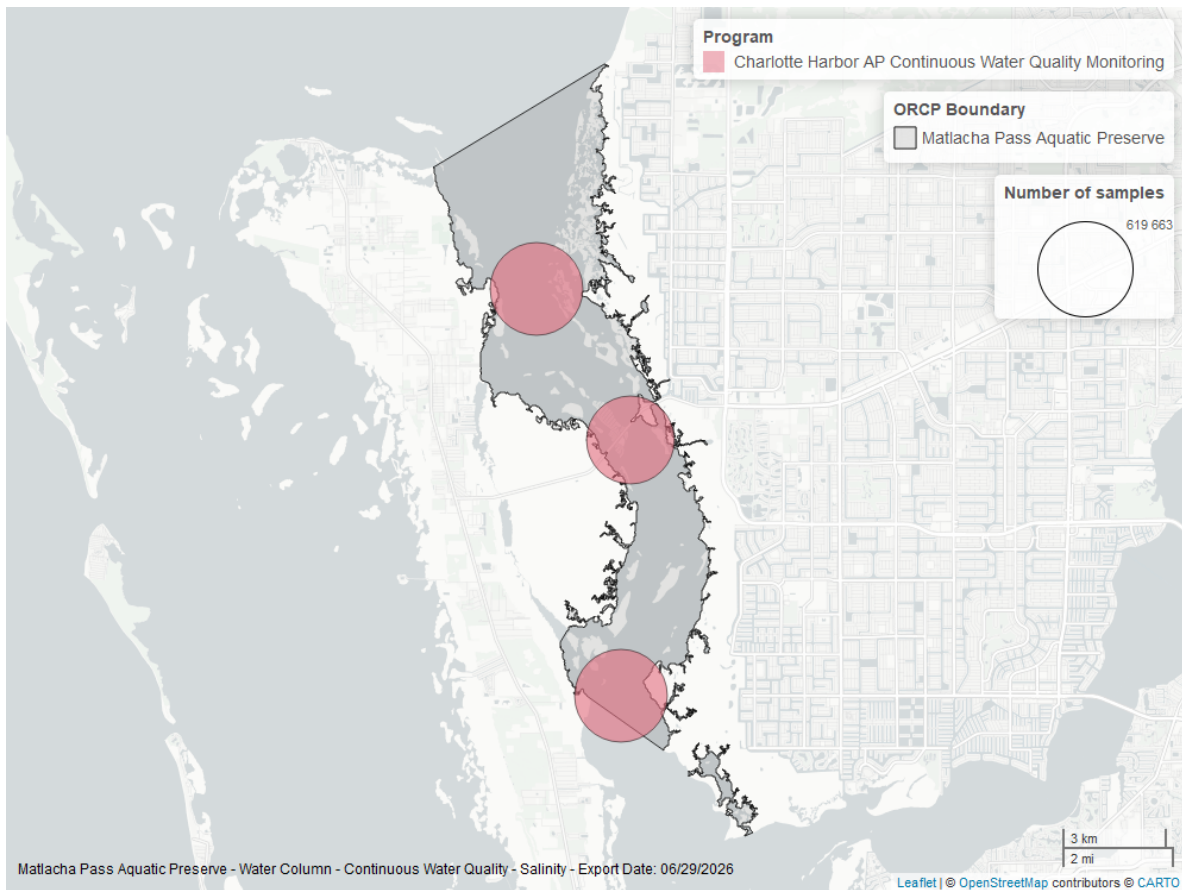


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

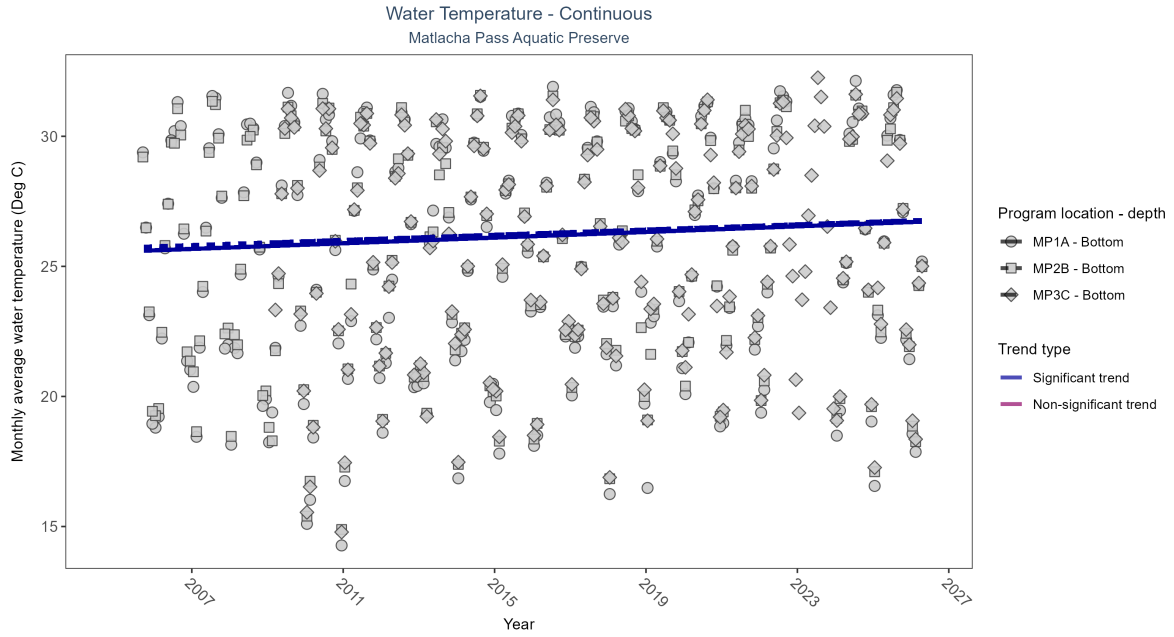


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At three program locations, monthly average water temperature increased between 0.05 and 0.05Deg C per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Increasing trend	653725	21	2005 - 2026	26.3	0.24	25.56	0.05	0
MP2B	Increasing trend	649187	21	2005 - 2026	26.1	0.20	25.71	0.05	0
MP3C	Increasing trend	567730	18	2009 - 2026	26.5	0.18	25.91	0.05	0

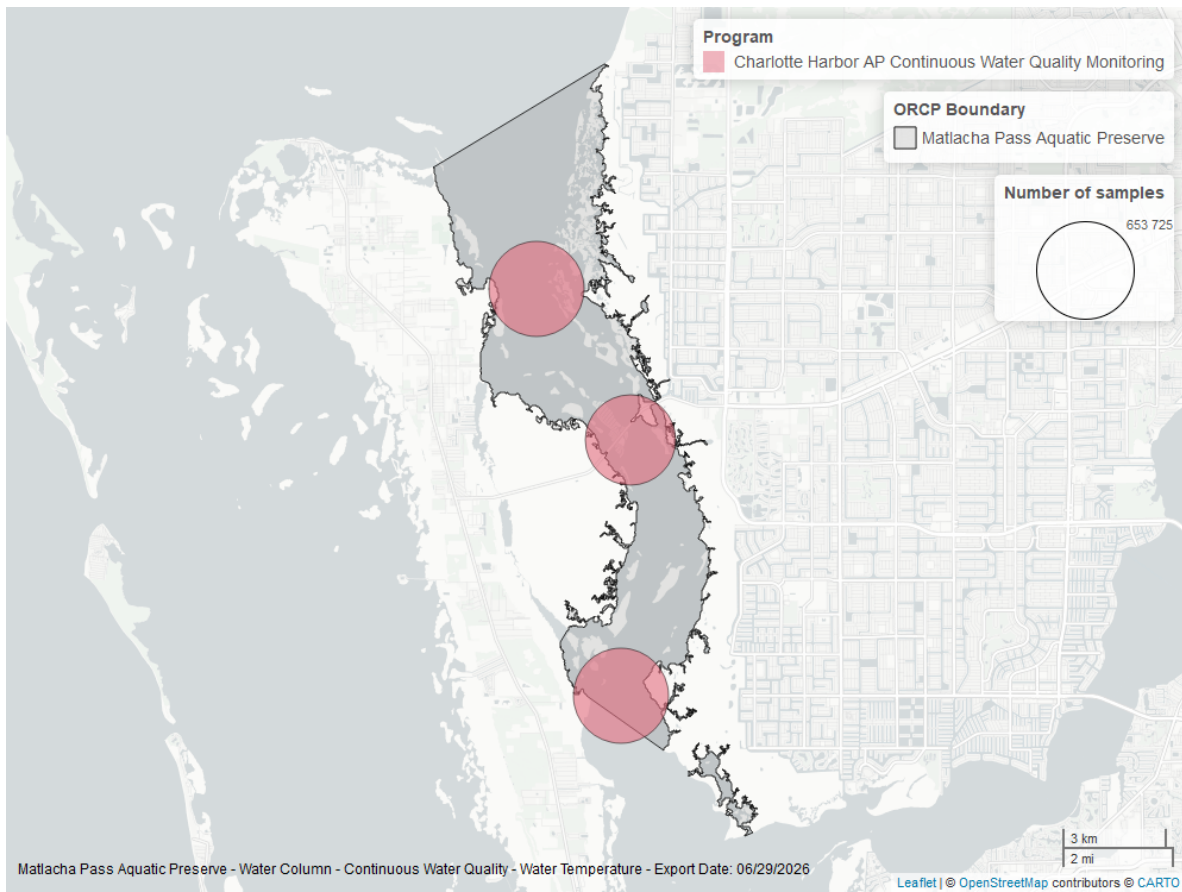


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

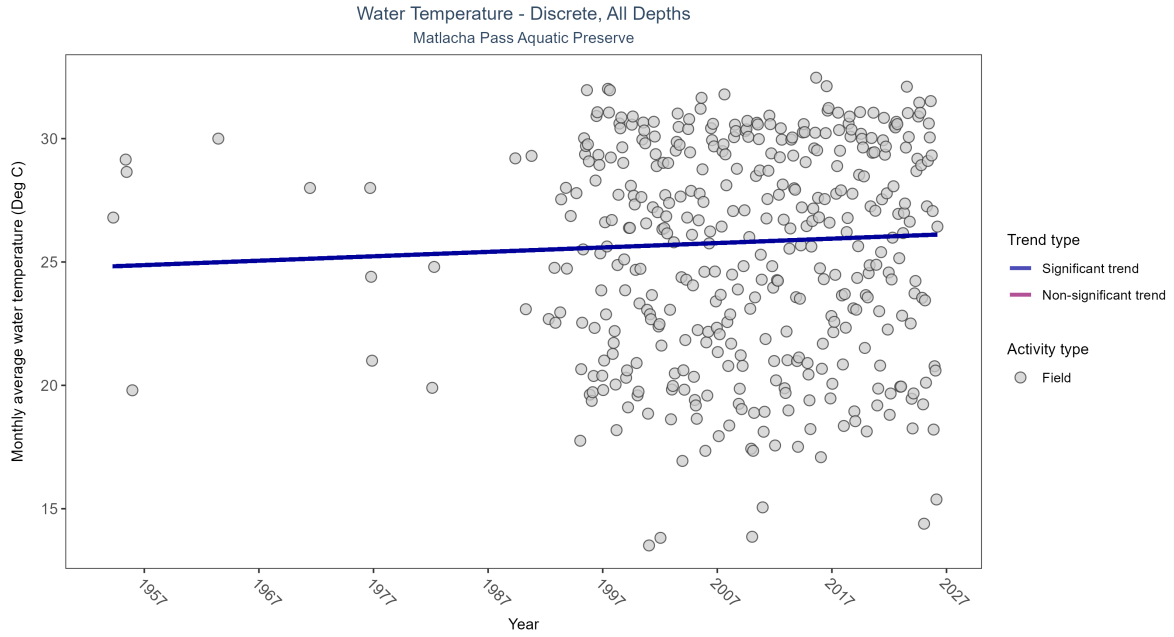


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 9: Monthly average water temperature increased by 0.02Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	12193	43	1954 - 2026	26.2	0.08788	24.81968	0.01788	0.0108

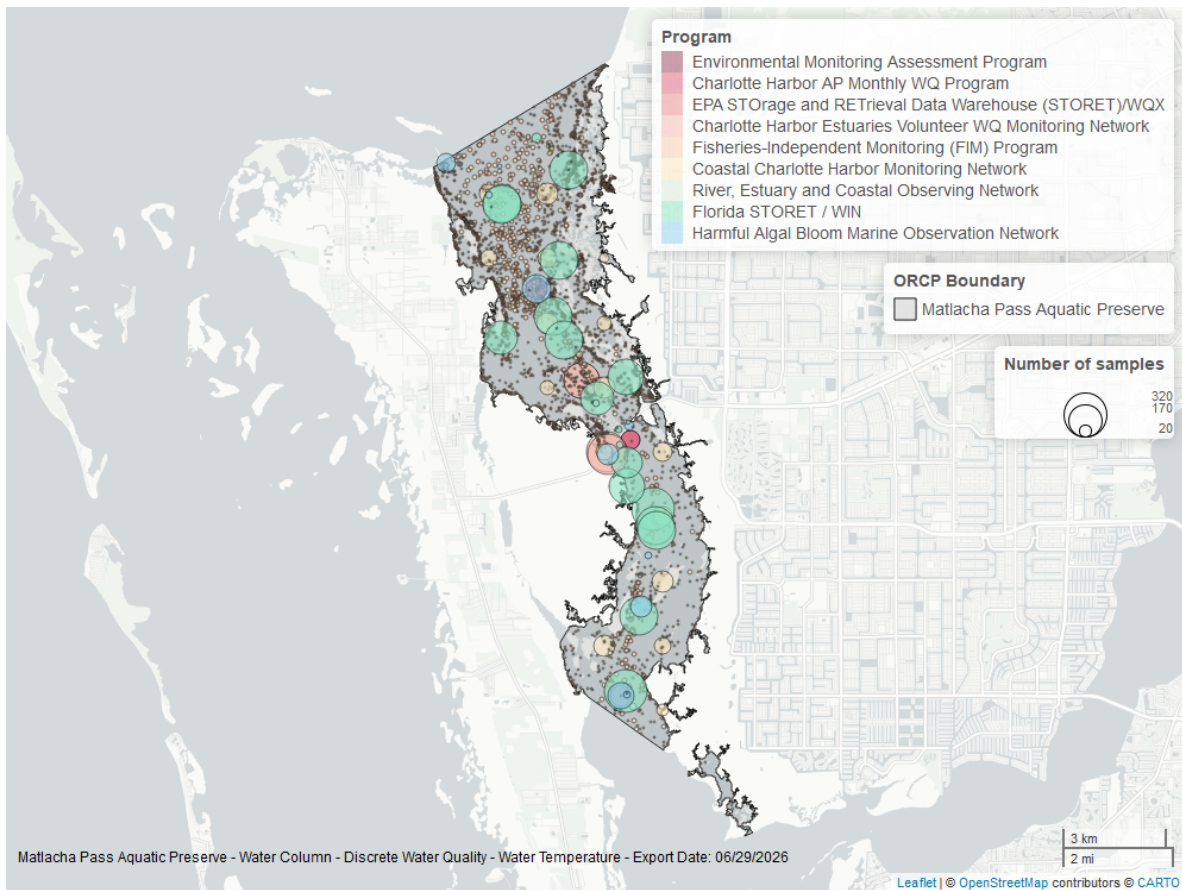


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

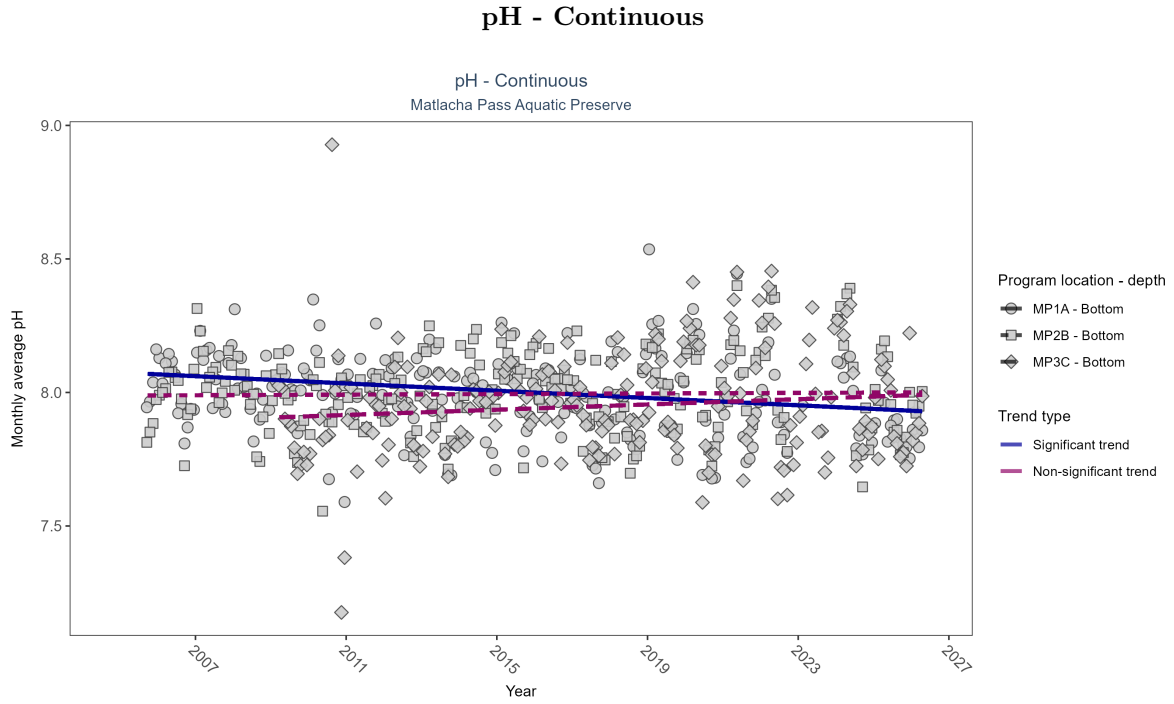


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At one program location, monthly average pH decreased by 0.01 pH units per year. No detectable change in monthly average pH was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Decreasing trend	566281	21	2005 - 2026	8.0	-0.22	8.07	-0.01	0.00
MP2B	No detectable trend	583207	21	2005 - 2026	8.0	0.01	7.99	0.00	0.79
MP3C	No detectable trend	504547	18	2009 - 2026	7.9	0.10	7.91	0.00	0.06

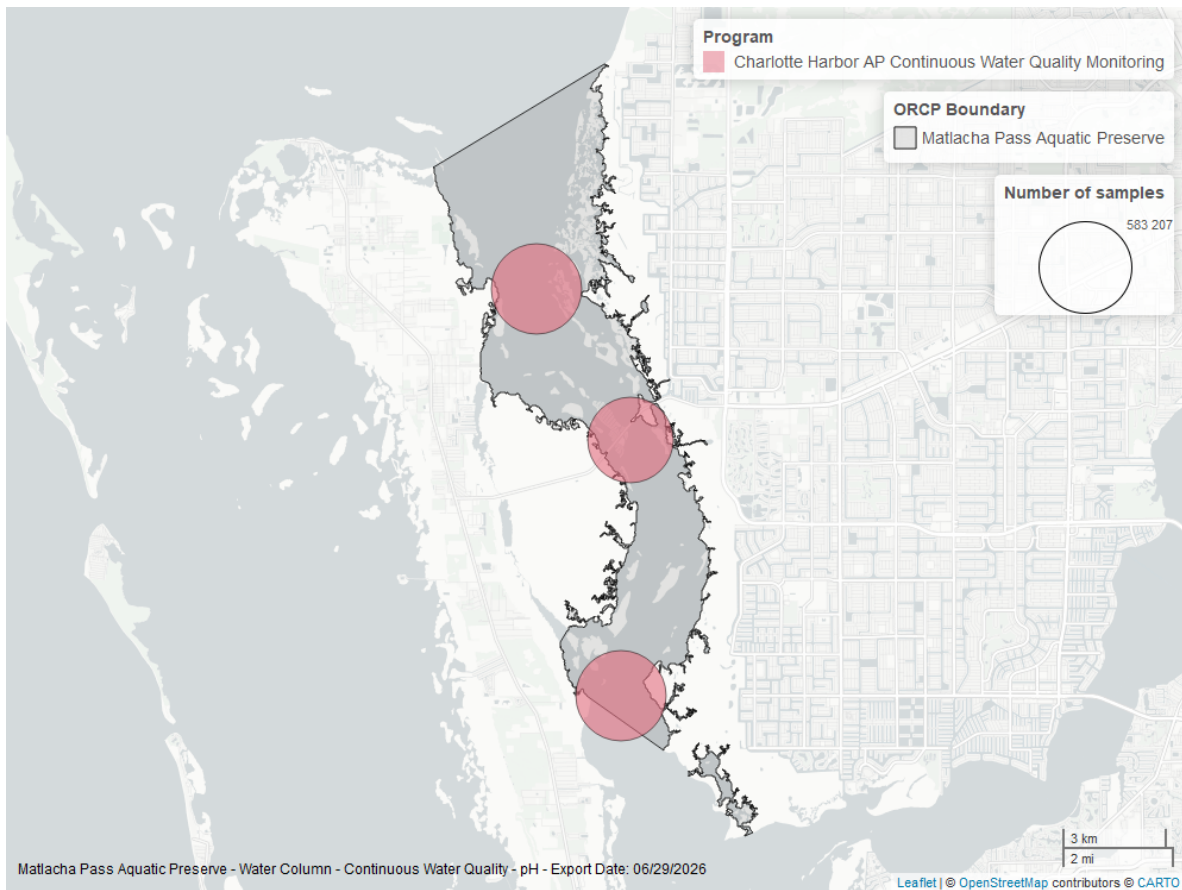


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

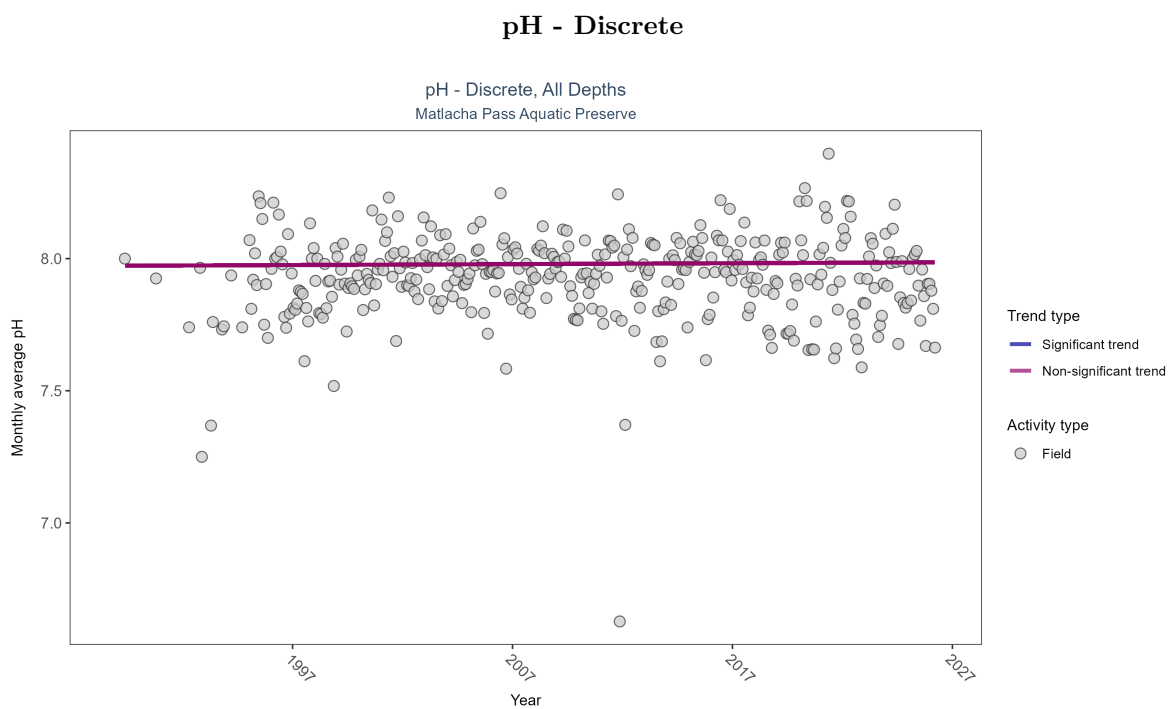


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 11: pH showed no detectable trend between 1989 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	11261	37	1989 - 2026	7.93	0.01125	7.97284	0.00035	0.6786

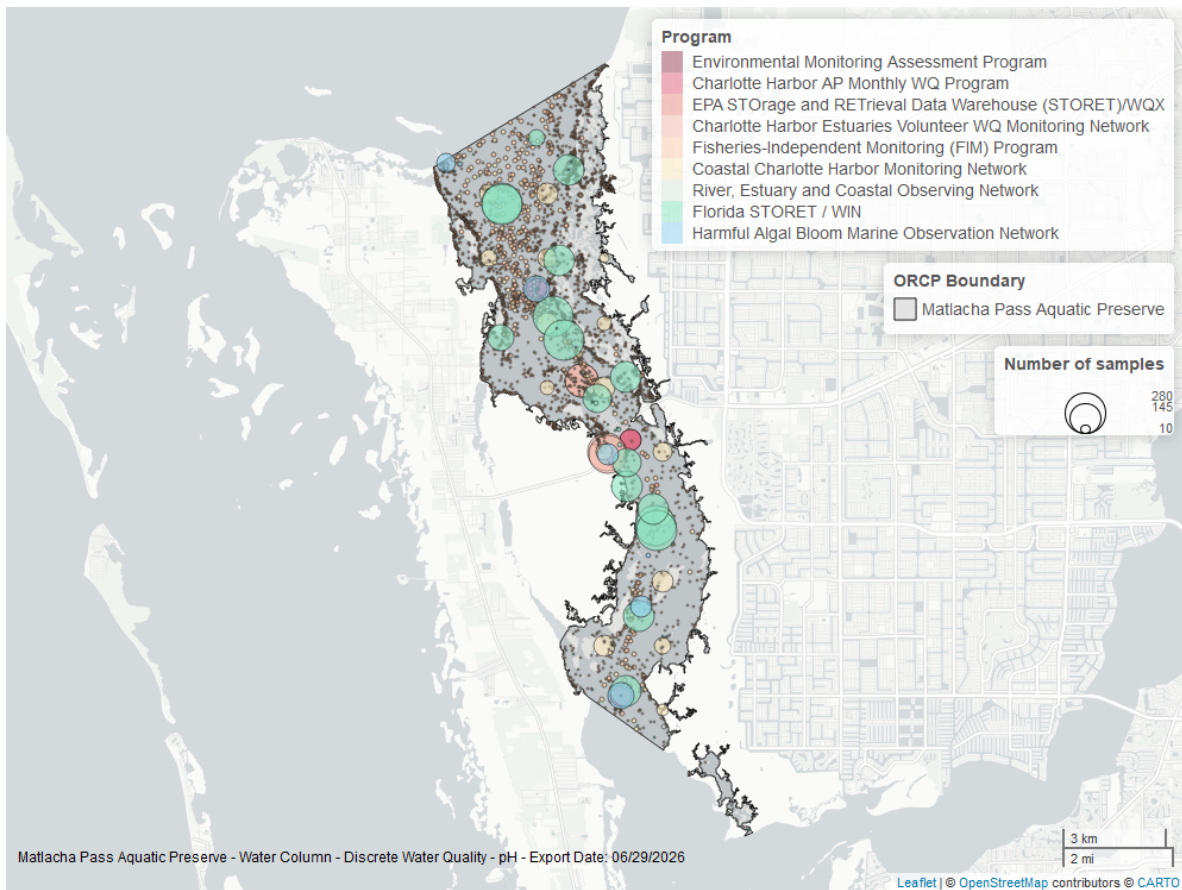


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

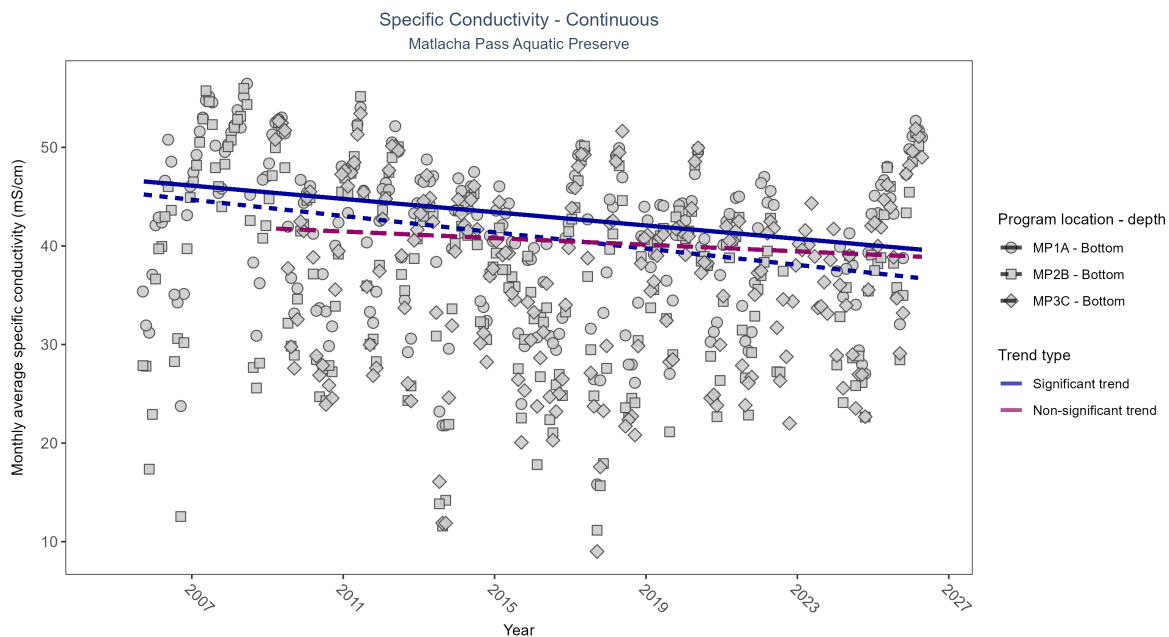


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At two program locations, monthly average specific conductivity decreased by 0.34 mS/cm per year at one site and by 0.41 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Decreasing trend	608827	21	2005 - 2026	42.33	-0.25	46.79	-0.34	0.00
MP2B	Decreasing trend	619663	21	2005 - 2026	38.62	-0.25	45.52	-0.41	0.00
MP3C	No detectable trend	548673	18	2009 - 2026	37.73	-0.07	41.80	-0.17	0.16

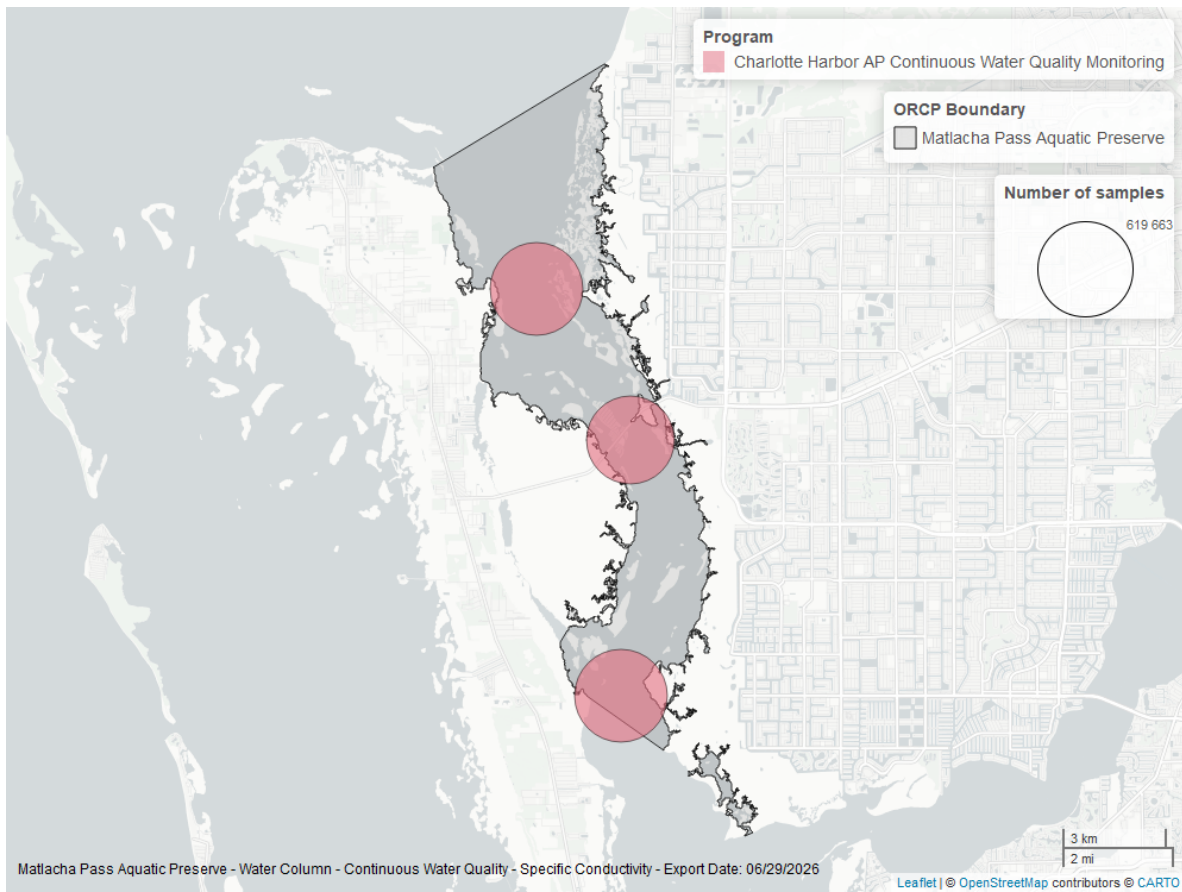


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

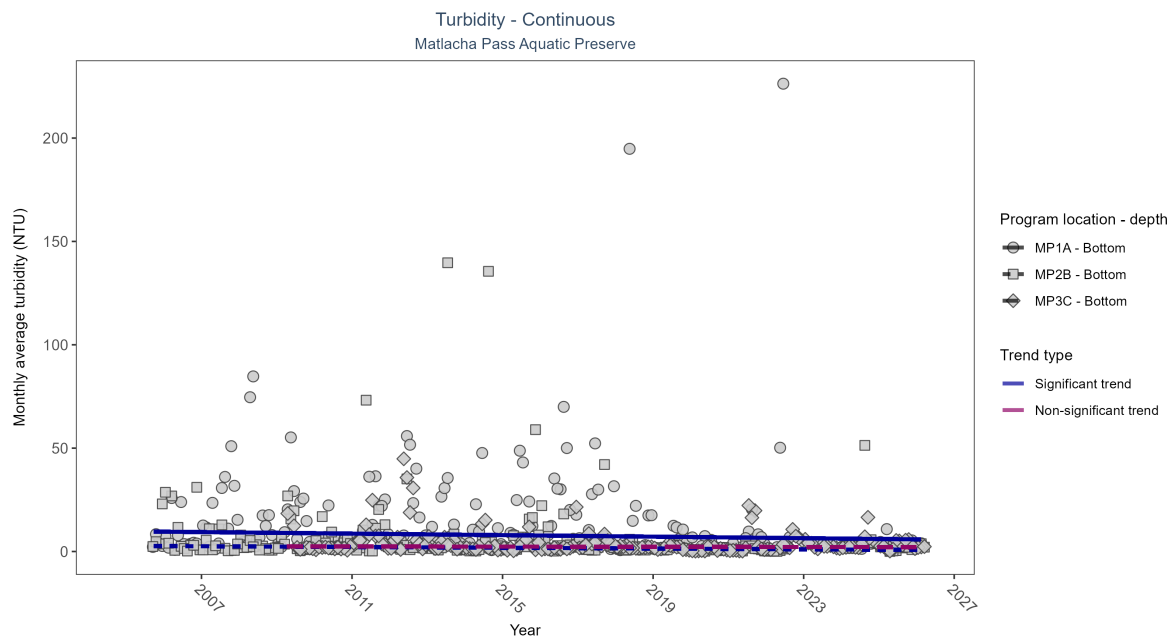


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At two program locations, monthly average turbidity decreased by 0.09 NTU per year at one site and by 0.18 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Decreasing trend	480933	21	2005 - 2026	2	-0.18	9.76	-0.18	0.00
MP2B	Decreasing trend	541889	21	2005 - 2026	1	-0.18	2.73	-0.09	0.00
MP3C	No detectable trend	472648	18	2009 - 2026	2	-0.04	2.49	-0.02	0.42

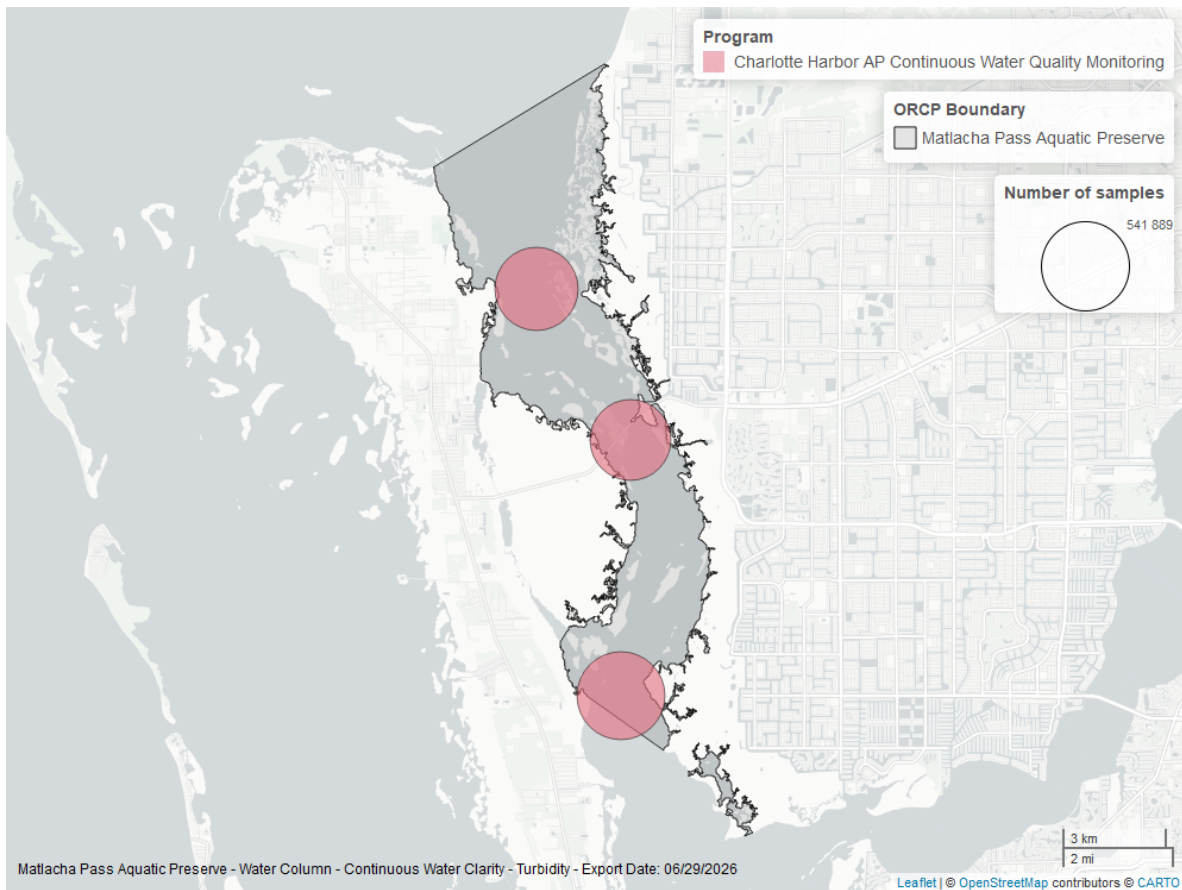


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

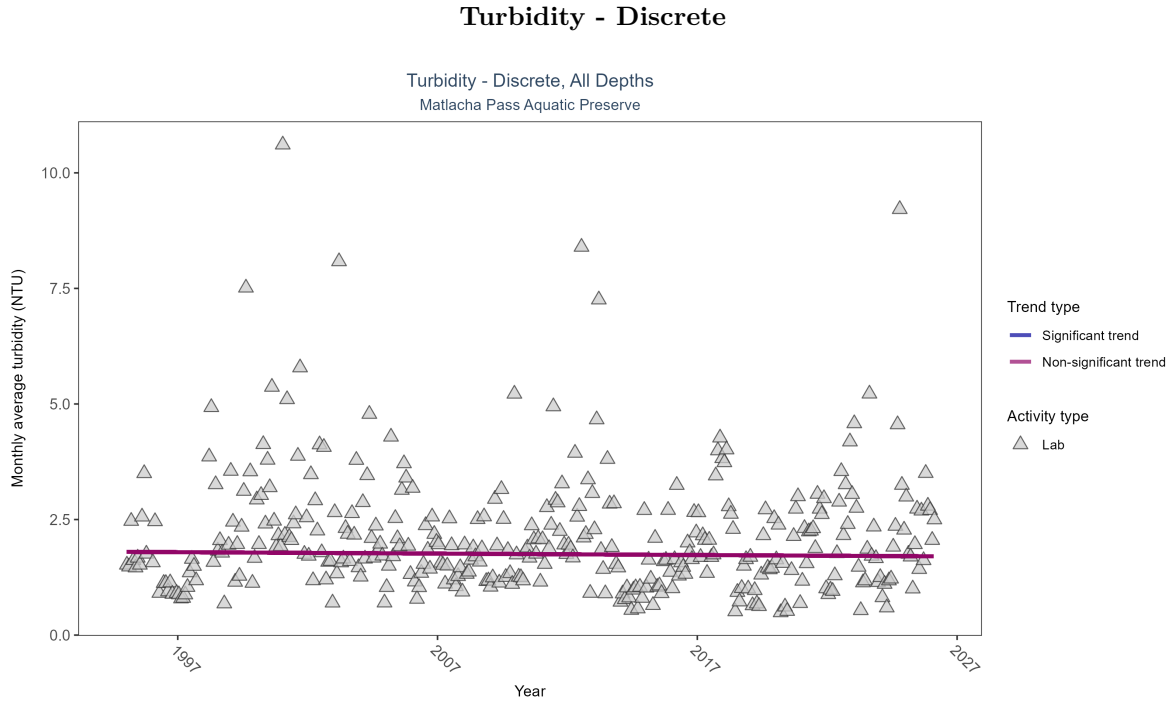


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 14: Turbidity showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3572	32	1995 - 2026	1.7	-0.01741	1.80022	-0.00308	0.5649

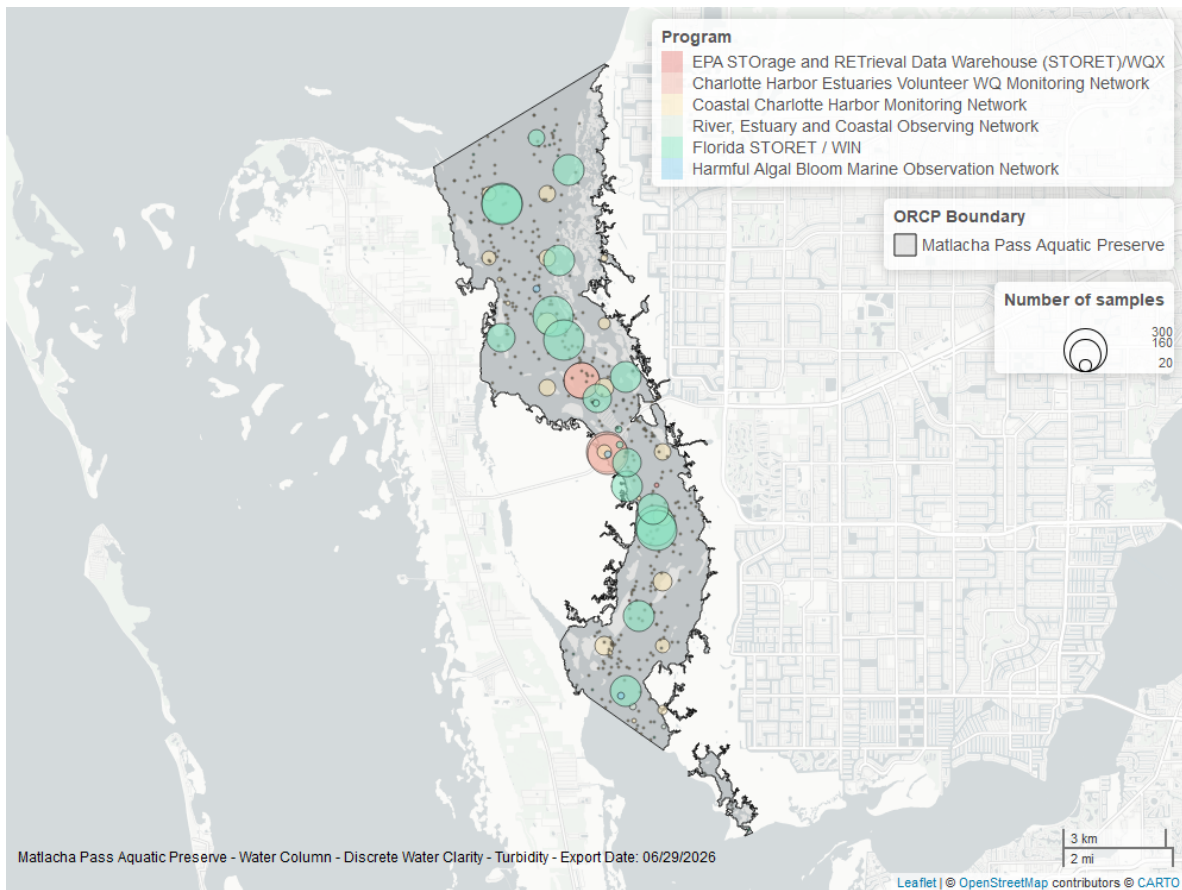


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

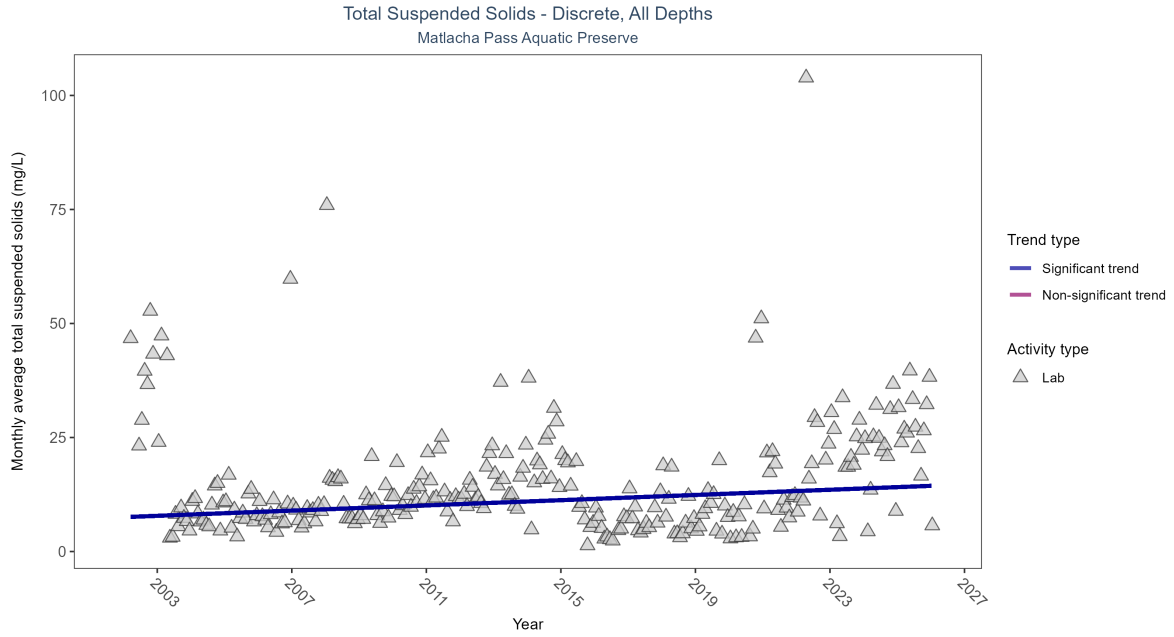


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 15: Monthly average total suspended solids increased by 0.29 mg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1936	25	2002 - 2026	10	0.14373	7.53926	0.28591	0.001

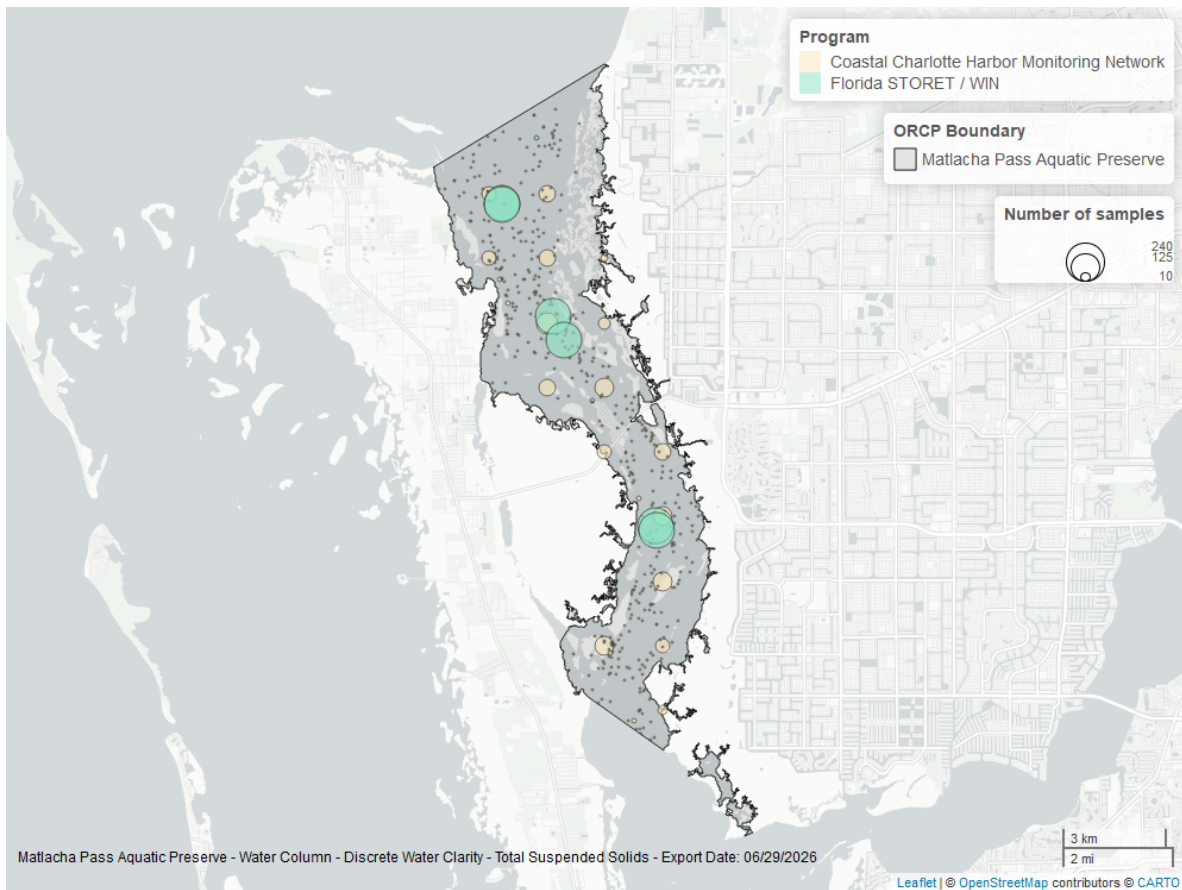


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

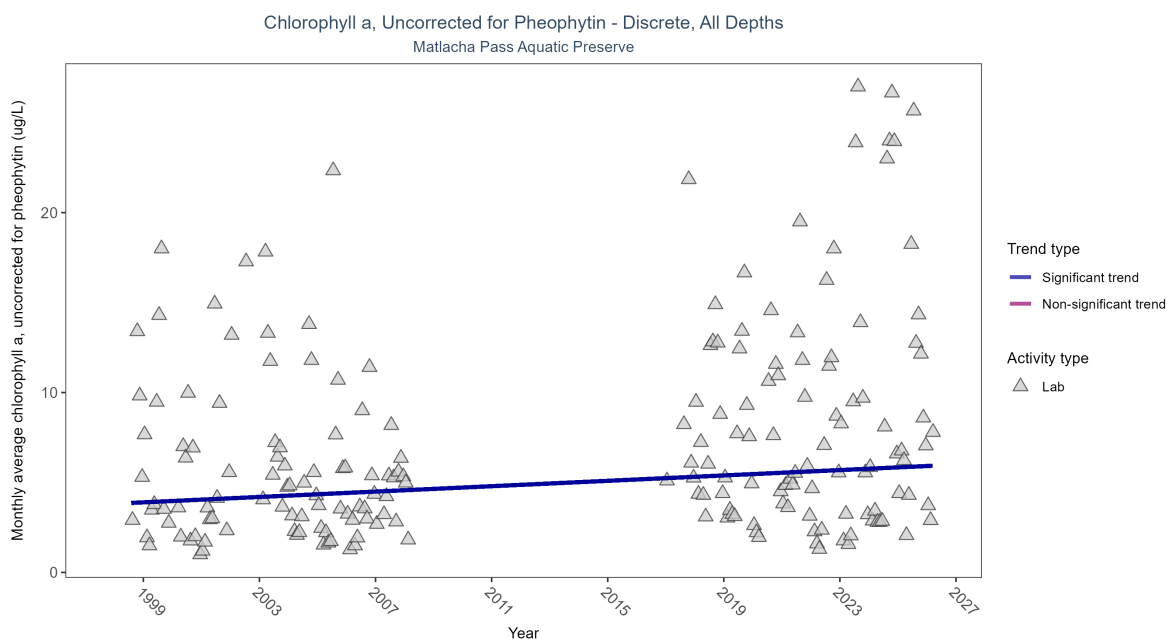


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.07 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	589	21	1998 - 2026	4.9	0.19982	3.82089	0.07476	0.0001

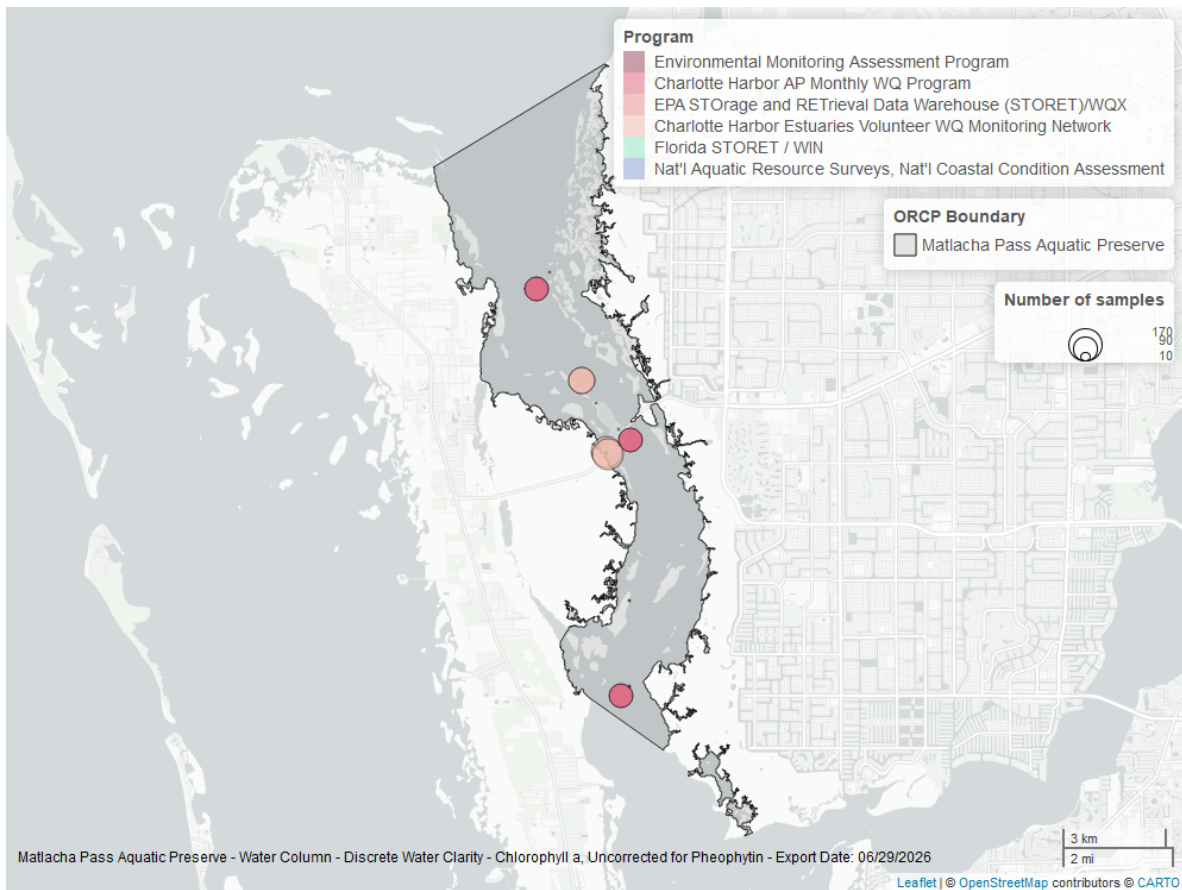


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

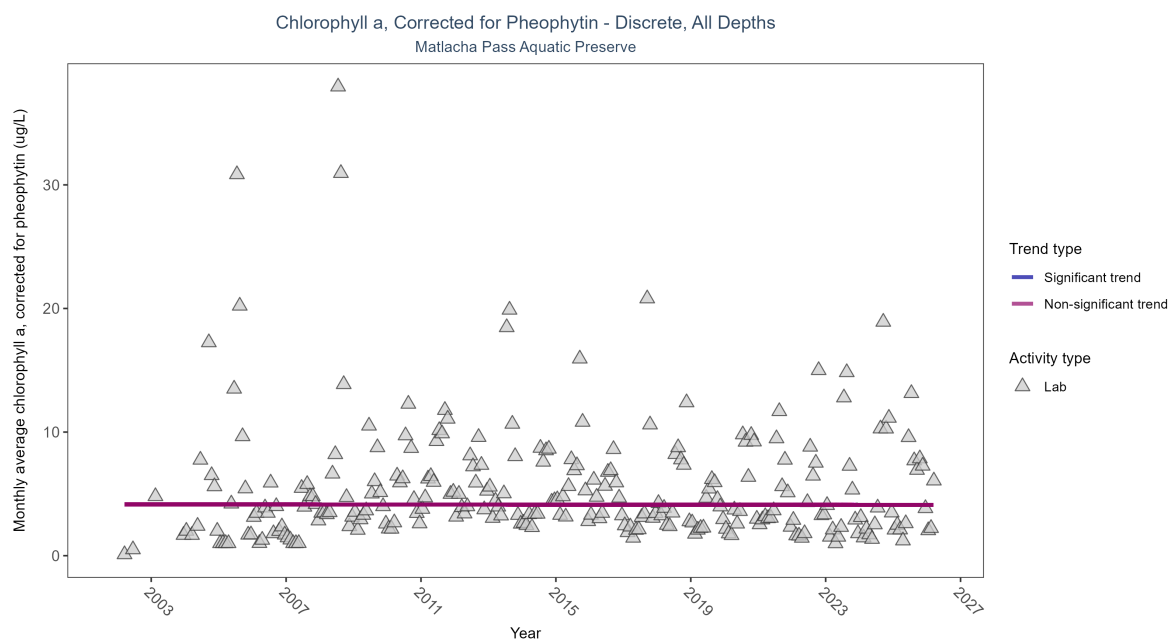


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2033	25	2002 - 2026	3.4	-0.00869	4.15204	-0.00188	0.8794

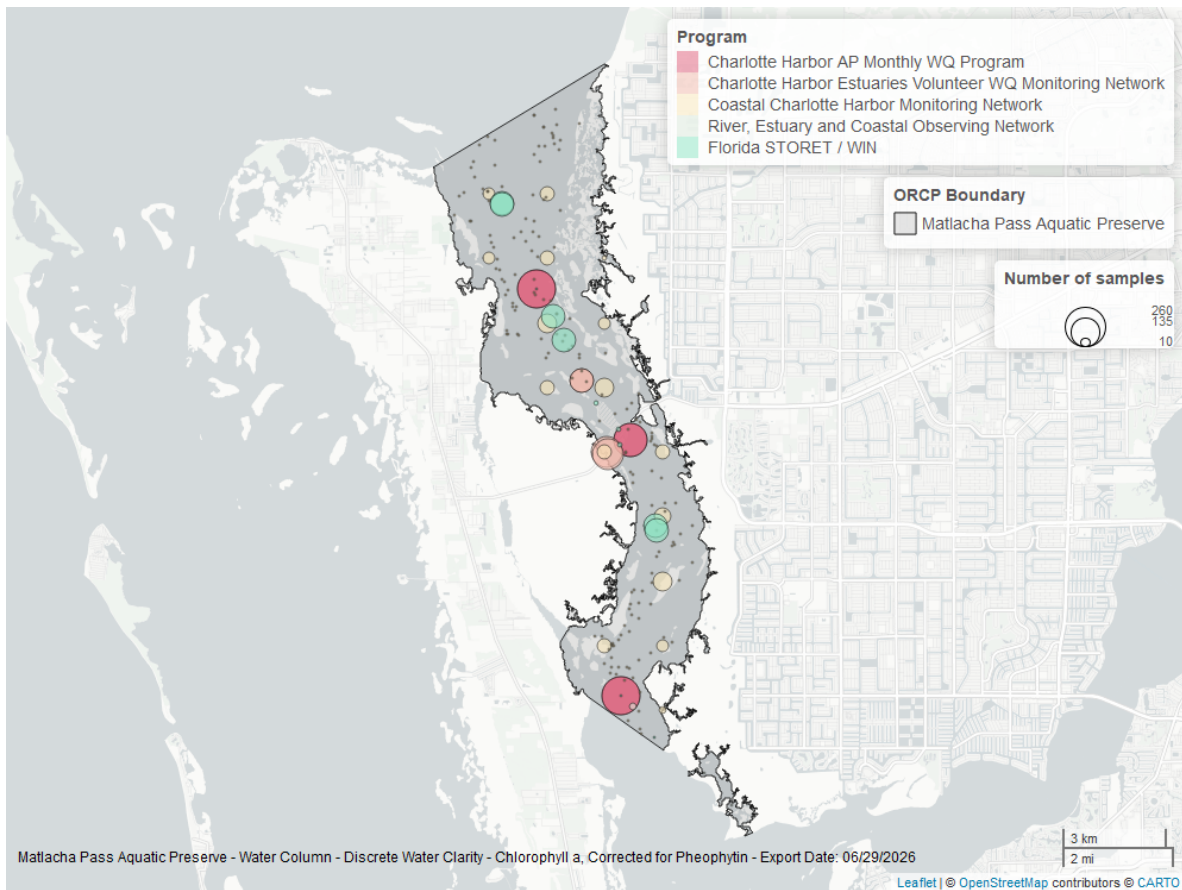


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

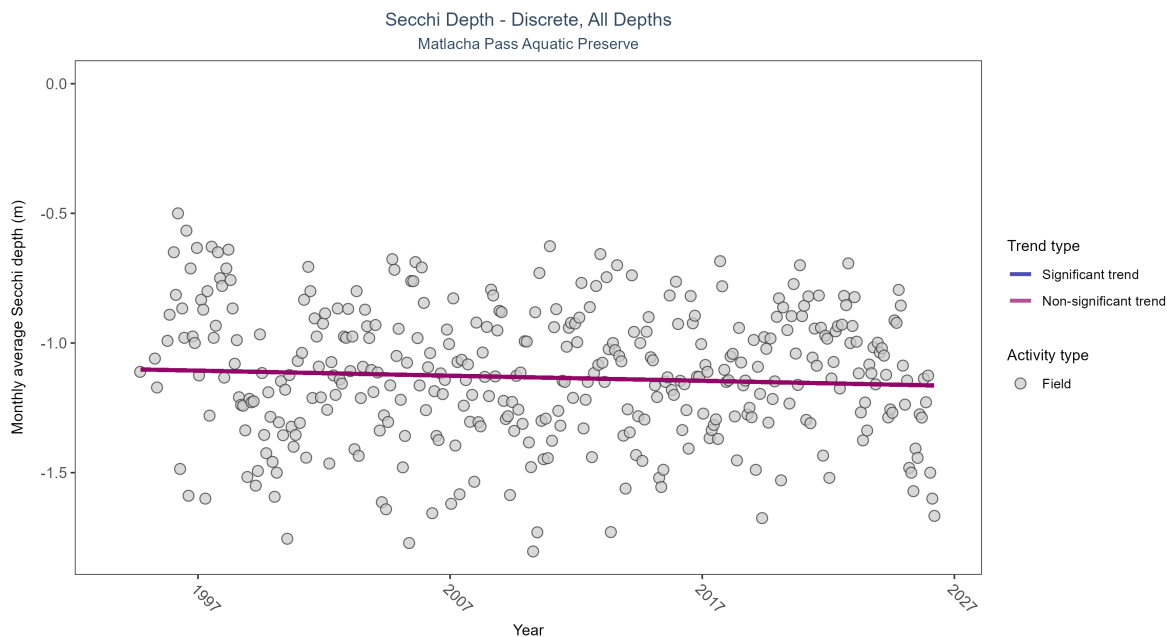


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: Secchi depth showed no detectable trend between 1994 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	6915	33	1994 - 2026	-1	-0.05054	-1.1006	-0.00196	0.1719

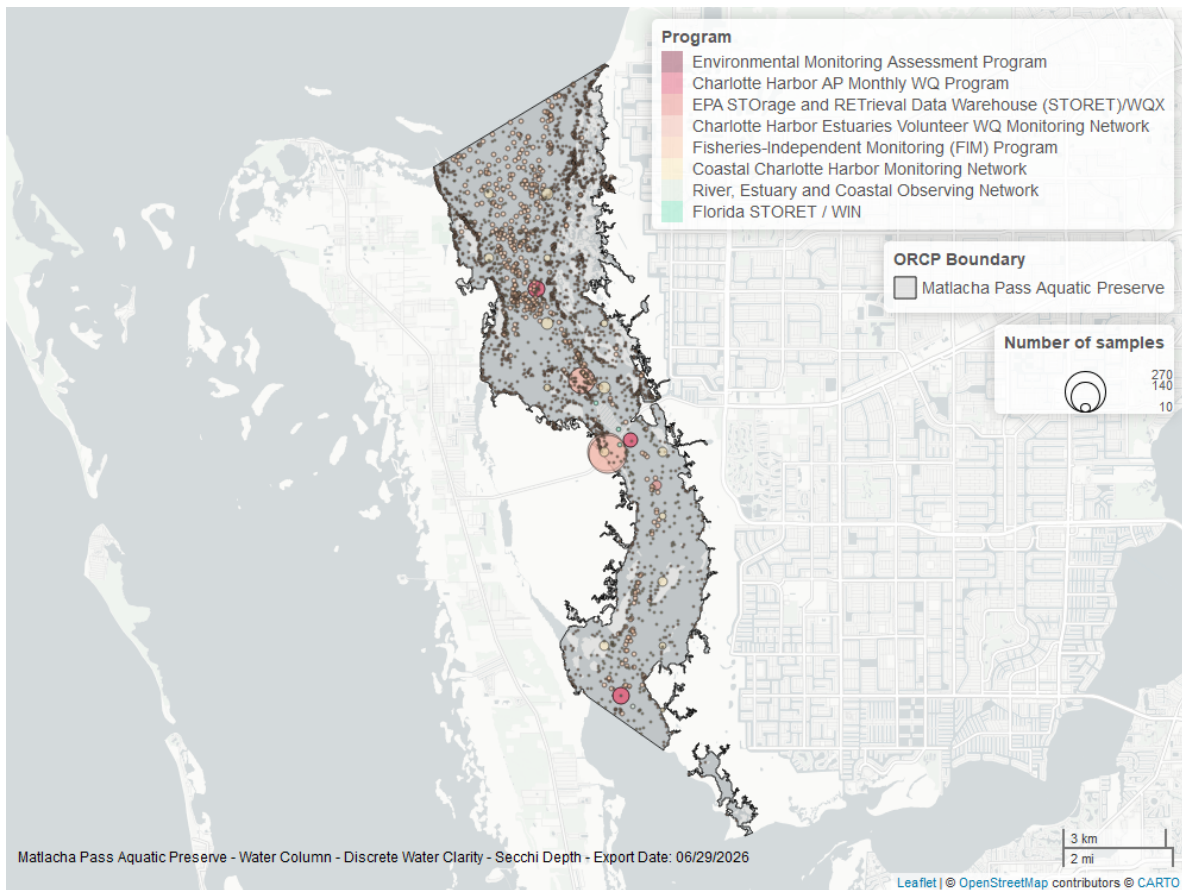


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

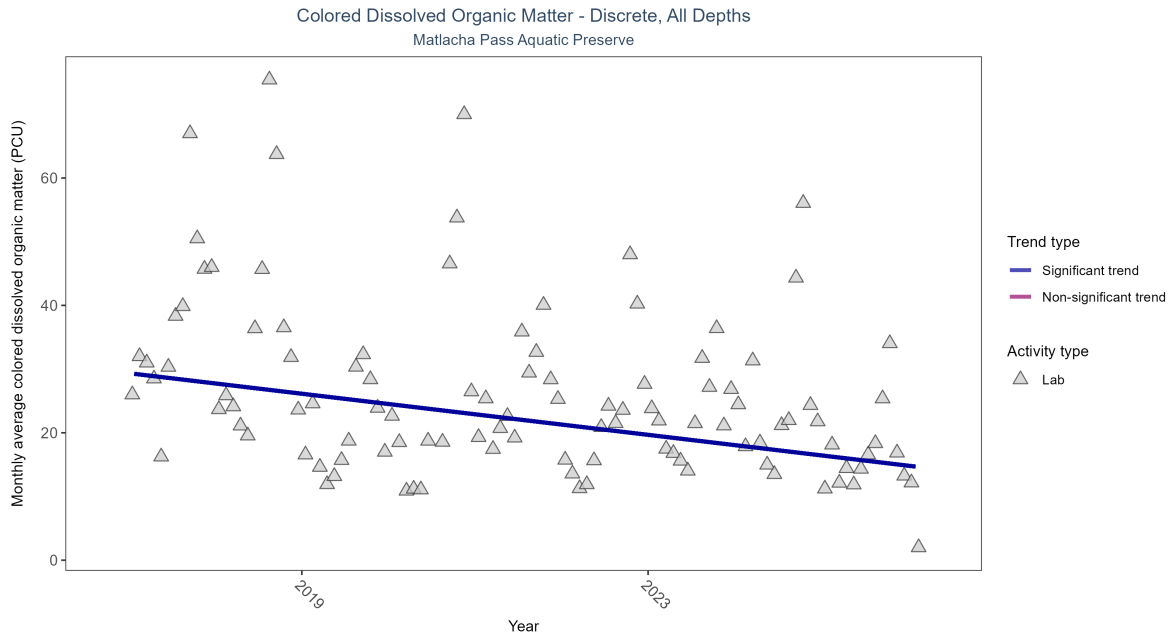


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 19: Monthly average colored dissolved organic matter decreased by 1.61 PCU per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	557	10	2017 - 2026	24	-0.40192	29.35809	-1.61338	0

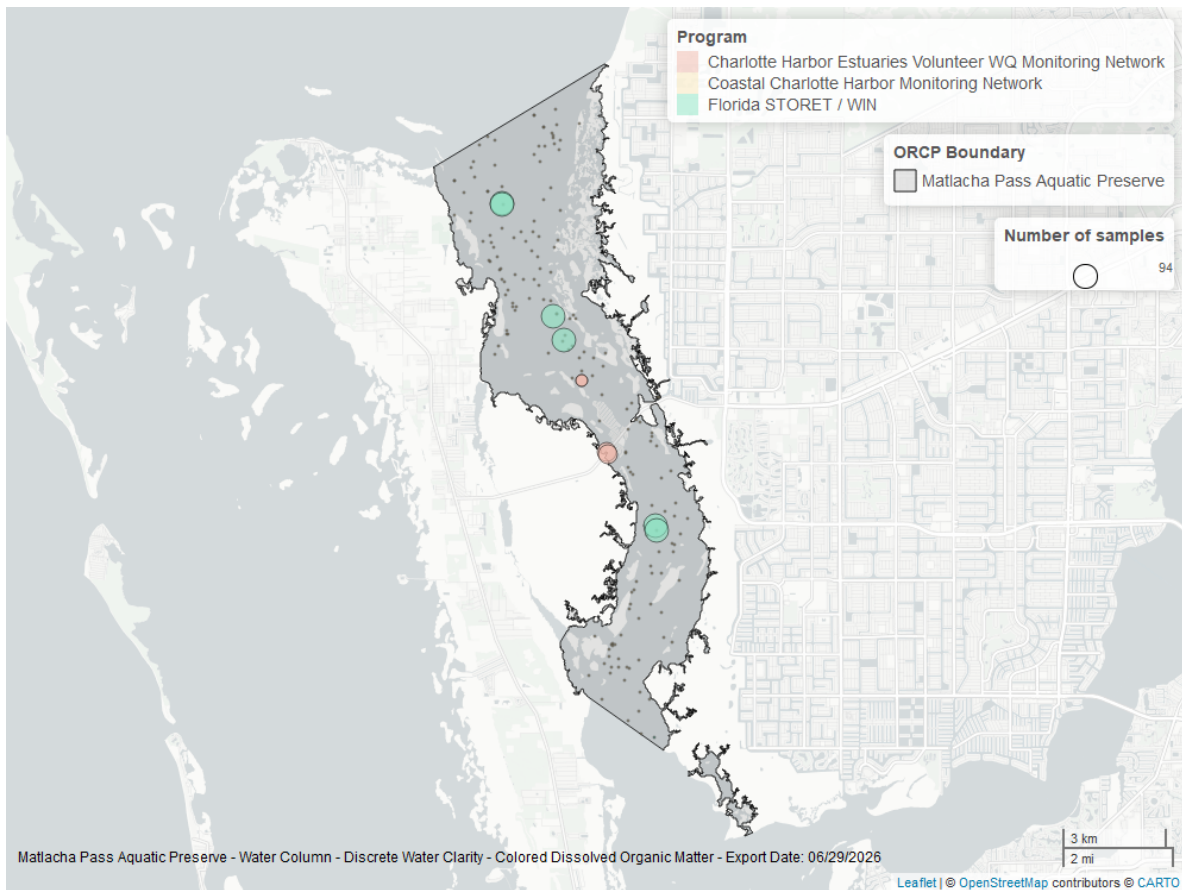


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.